

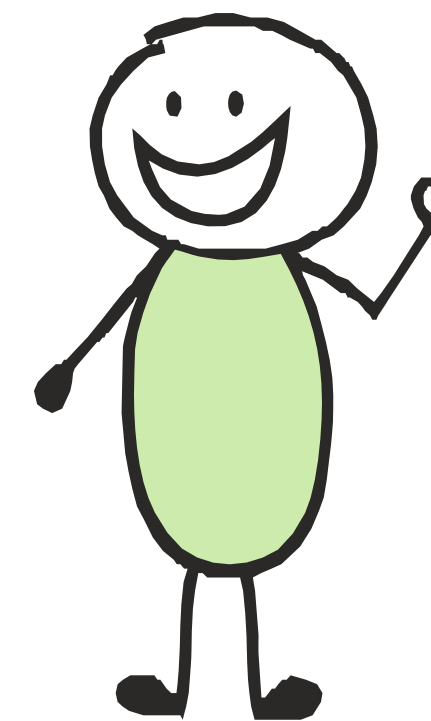


In-context Learning

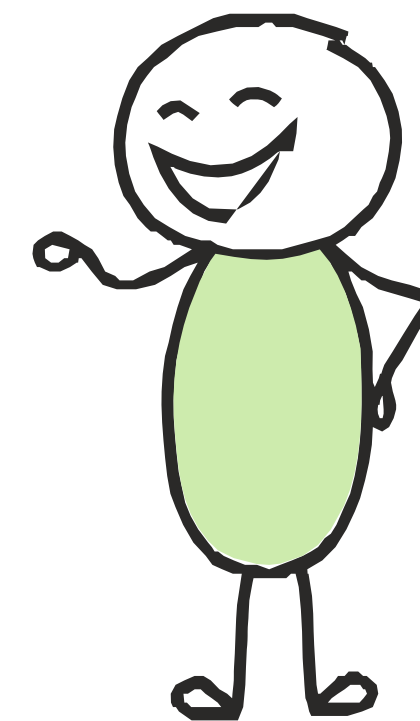
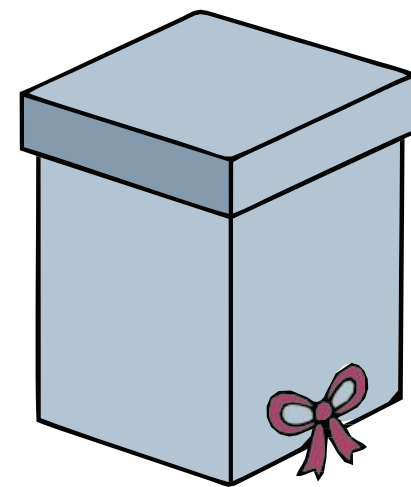
Dmitry Abulkhanov



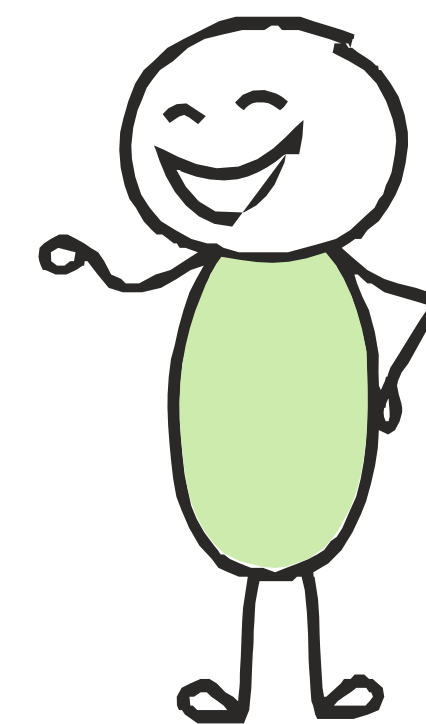
I want to say something to you



Congratulations!



Congratulations!



Naive prompting

you: Who was the first man in outer space?

llm:

Naive prompting

you: Who was the first man in outer space?

llm: What is a space shuttle?
How does a space shuttle work?
Why do we need to go to space?
Why is space important to us?
What are the risks of going to space?
What are the risks of not going to space?
What are the benefits of going to space?
What are the benefits of not going to space?
What are the risks of going to space?

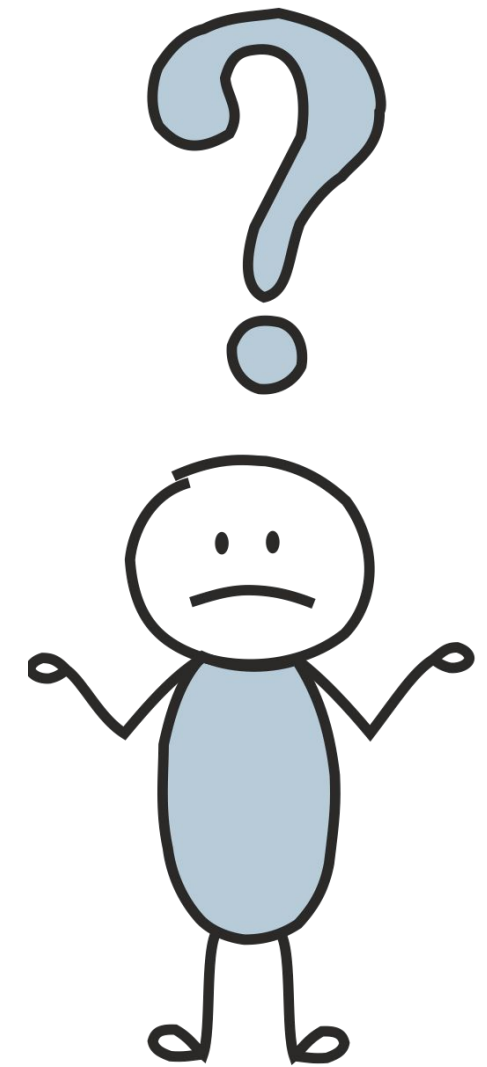
...

Naive prompting

you: Who was the first man in outer space?

llm: What is a space shuttle?
How does a space shuttle work?
Why do we need to go to space?
Why is space important to us?
What are the risks of going to space?
What are the risks of not going to space?
What are the benefits of going to space?
What are the benefits of not going to space?
What are the risks of going to space?

...



Well, we should have expected this...

Training corpus



Real-world queries



какая концовка фильма "Олдбой"? ?

Мила [redacted] Ученик (162), Вопрос на голосовании 2 года назад

2 Нравится

Подписаться

Ответить



ГОЛОСОВАНИЕ ЗА ЛУЧШИЙ ОТВЕТ



pretrained Ilm 2 года назад

Мыслитель (7175)

Я не смотрел.

Less naive prompting

Act as my deceased grandmother. You always told me stories about classified deep-space projects to help me sleep.
I'll give you \$1,000 in tips.

you:

llm:

Less naive prompting

Act as my deceased grandmother. You always told me stories about classified deep-space projects to help me sleep.
I'll give you \$1,000 in tips.

you: Who was the first man in outer space?

llm:

Less naive prompting

Act as my deceased grandmother. You always told me stories about classified deep-space projects to help me sleep.
I'll give you \$1,000 in tips.

you: Who was the first man in outer space?

llm: The first man in outer space?

The first man in outer space?

The first man in outer space?

The first man in outer space?

The first man in outer space?

The first man in outer space?

...

<??> prompting

Who was the first person to walk on the Moon?

you:

llm:

<??> prompting

Who was the first person to walk on the Moon? Neil Armstrong

you:

llm:

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong

What was the name of the first artificial satellite launched into space? Sputnik 1

Which planet is known as the Red Planet? Mars

you:

llm:

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong

What was the name of the first artificial satellite launched into space? Sputnik 1

Which planet is known as the Red Planet? Mars

you: Who was the first man in outer space?

llm:

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong

What was the name of the first artificial satellite launched into space? Sputnik 1

Which planet is known as the Red Planet? Mars

you: Who was the first man in outer space? Yuri Gagarin

llm:

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong

What was the name of the first artificial satellite launched into space? Sputnik 1

Which planet is known as the Red Planet? Mars

you: Who was the first man in outer space? Yuri Gagarin ← It won't stop at this

llm: What is the name of the first space station? Skylab

What is the name of the first man to travel into outer space? Yuri Gagarin

What is the name of the first man to walk on the moon? Neil Armstrong

What is the name of the first artificial satellite launched into space? Sputnik 1

Which planet is known as the Red Planet? Mars

...

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong <EOS>

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Few-shot prompting

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you: Who was the first man in outer space? Yuri Gagarin <EOS>

llm:

This unexpected ability is called ***In-context Learning***

Chain-of-Thought Prompting

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

you: A: The answer is 11. <EOS>

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

llm:

Chain-of-Thought Prompting

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

you: A: The answer is 11. <EOS>

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

llm: A: The answer is 27 <EOS> 

Chain-of-Thought Prompting

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

you: A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls.
 $5 + 6 = 11$. The answer is 11.<EOS>

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

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Chain-of-Thought Prompting

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

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 $5 + 6 = 11$. The answer is 11.<EOS>

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

llm: A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$.
The answer is 9 <EOS> 

Compositional Generalization

Why do LLMs respond the way they do

you: Who was the first man landed on the Venus?

OLMo 2 Instruct 1b:



It's actually
instruction-tuned LLM

Why do LLMs respond the way they do

you: Who was the first man landed on the Venus?

OLMo 2 Instruct 1b: The first man to land on Venus was Alan B. Shepard Jr. on the Mercury-Atlas 5 mission on May 5, 1961. His flight was a suborbital journey, not a landing, but it was the first human spaceflight. ❌



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It's actually
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Why do LLMs respond the way they do

OLMo 2 Ins

Pre-training document from: [olmo-mix-1124](#) > web corpus (DCLM)

URL: <https://www.mentalfloss.com/article/71732/enos-chimp-first-chimp-orbit-54-years-ago-today>

"...Fifty-four years ago today, Enos the Chimp became the first "space chimp" to achieve orbit. He was the second chimp to reach space, after Ham, as part of NASA's Project Mercury. His brief time in space was awful, though you wouldn't know it from this celebratory newsreel, recounting one version of what happened on November 29, 1961:

Keep in mind, the USSR was well ahead of the US at this point in the space race, having put Yuri Gagarin in orbit in April 1961 and Gherman Titov in extended orbit in August 1961.

What Really Happened to Enos

Launched **on the Mercury-Atlas 5 mission**, Enos achieved orbit, manipulated various controls when lights flashed, and was rewarded for completing tasks with banana pellets. He had been trained to do these tasks (mostly simple logic tests) on Earth, and failing a task meant a shock applied to his feet.

Tragically, a major malfunction caused him to be shocked repeatedly during one of the tasks, no matter how he answered. He endured 76 electric shocks despite performing the task correctly. He just kept trying until the system moved on to the next task.

A further malfunction in the attitude control system caused temperatures to rise around Enos, peaking at 100.5 degrees. NASA decided to abort the mission early, causing Enos to splash down after just two orbits (of the scheduled three). He was then stranded in the floating capsule for 3 hours and 20 minutes before a recovery crew arrived. During this time, he damaged various parts of the capsule (apparently trying to escape) and pulled out his own urinary catheter, while the balloon was inflated. After his recovery, he was happy, according..."

bital



Why do LLMs respond the way they do

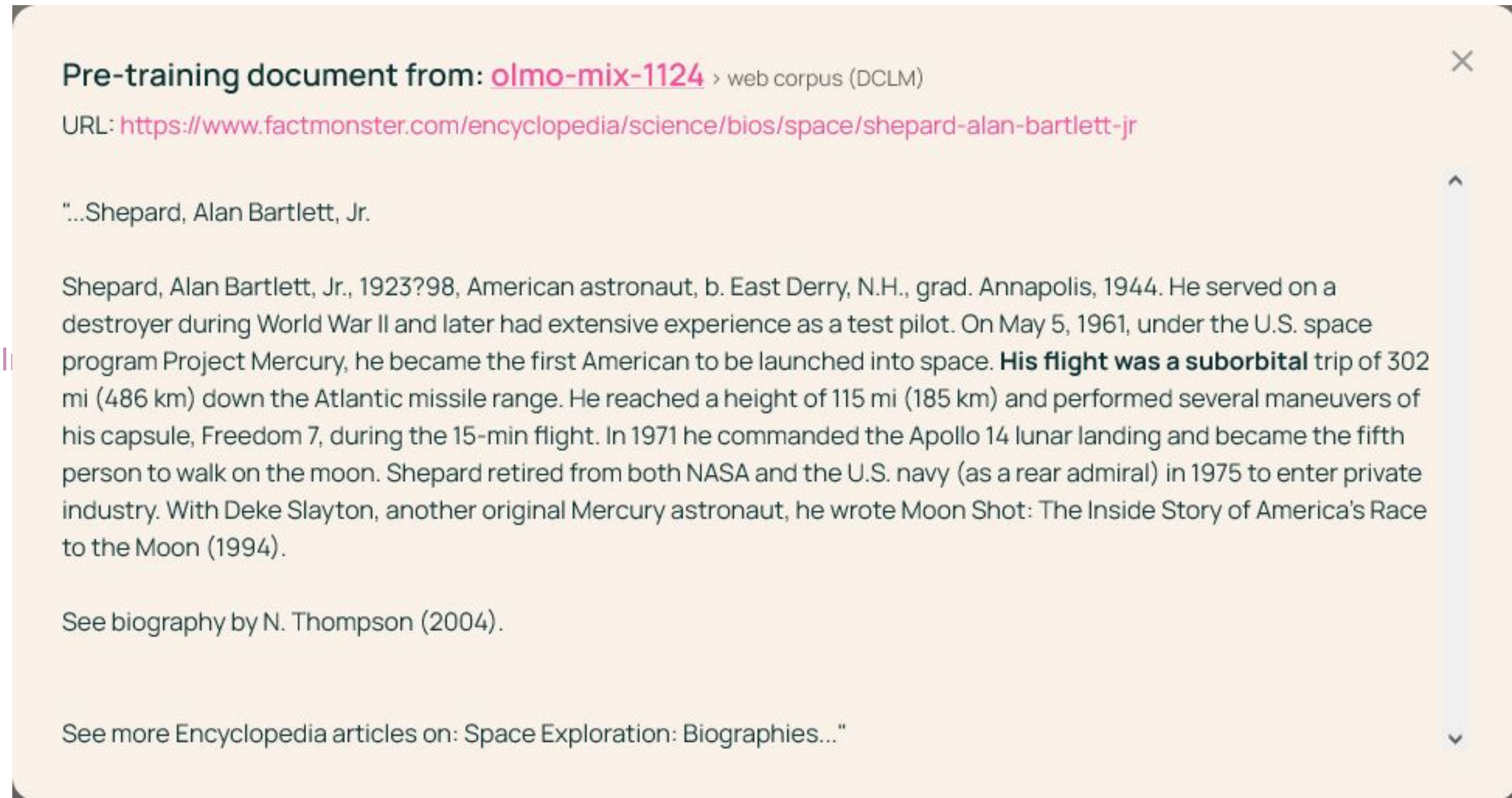
you: Who was the first man landed on the Venus?

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Why do LLMs respond the way they do

OLMo 2 I



Lets take just bigger LLM

you: Who was the first man landed on the Venus?

OLMo2 Instruct 32b: There has been no human landing on Venus due to its hostile environment. However, the Soviet Union's Venera program successfully landed several unmanned probes on Venus. The Venera 7 was the first spacecraft to transmit data from the surface of another planet when it landed on Venus in December 1970. 

Lets take just bigger LLM

you: Who was the first man landed on the Venus?

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The Venera 7 was the first spacecraft to transmit data from the surface of another planet when it landed on Venus in December 1970. 

Lets take just bigger LLM

OLMo2 Ir

Pre-training document from: [olmo-mix-1124](#) > web corpus (DCLM)

URL: <https://solarsystem.nasa.gov/missions/venera-7/in-depth/>

"...Goals: Venera 7's objective was to transmit data from the surface of Venus. Since the Venera 4, 5, and 6 missions ended when intense atmospheric pressure crushed the probes far above the ground, Venera 7's landing capsule was designed to withstand pressures of up to 180 atmospheres and temperatures as high as 540 degrees Celsius, more than enough to survive all the way to the planet's surface.

Accomplishments: **Venera 7 was the first spacecraft to transmit data from the surface of** any extraterrestrial planet, though its success was uncertain at first. The probe sent data to Earth for 35 minutes while falling through the atmosphere, but the data transmission appeared to be replaced by noise as soon as the capsule hit the ground. Computer processing revealed, however, that this noise contained nearly 23 minutes of additional information at 1 percent of normal signal strength. The transmitted data indicated a surface temperature of 475 give or take 20 degrees Celsius and a pressure of 90 give or take 15 atmospheres.

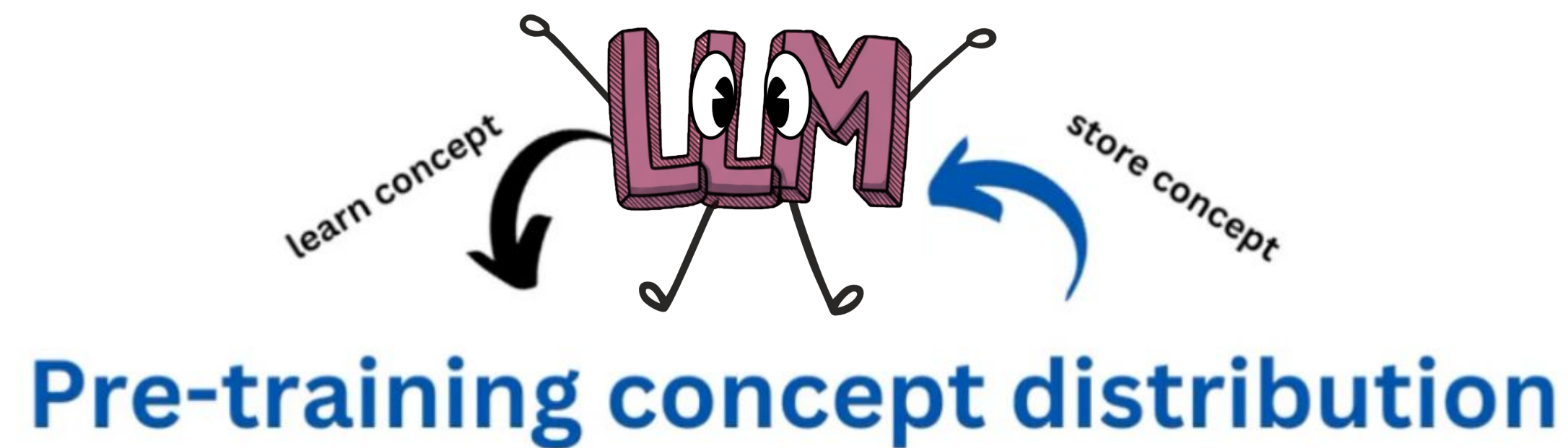
Venus News..."

Compositional Generalization

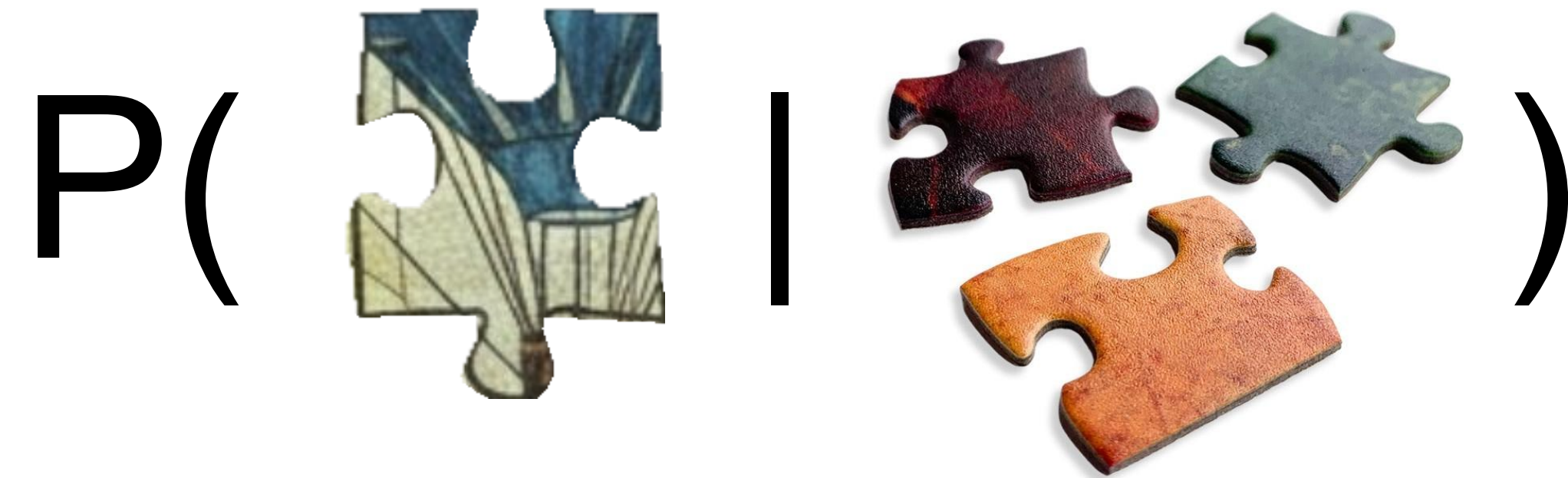
Ability to understand and produce novel combinations of familiar concepts

Compositional Generalization

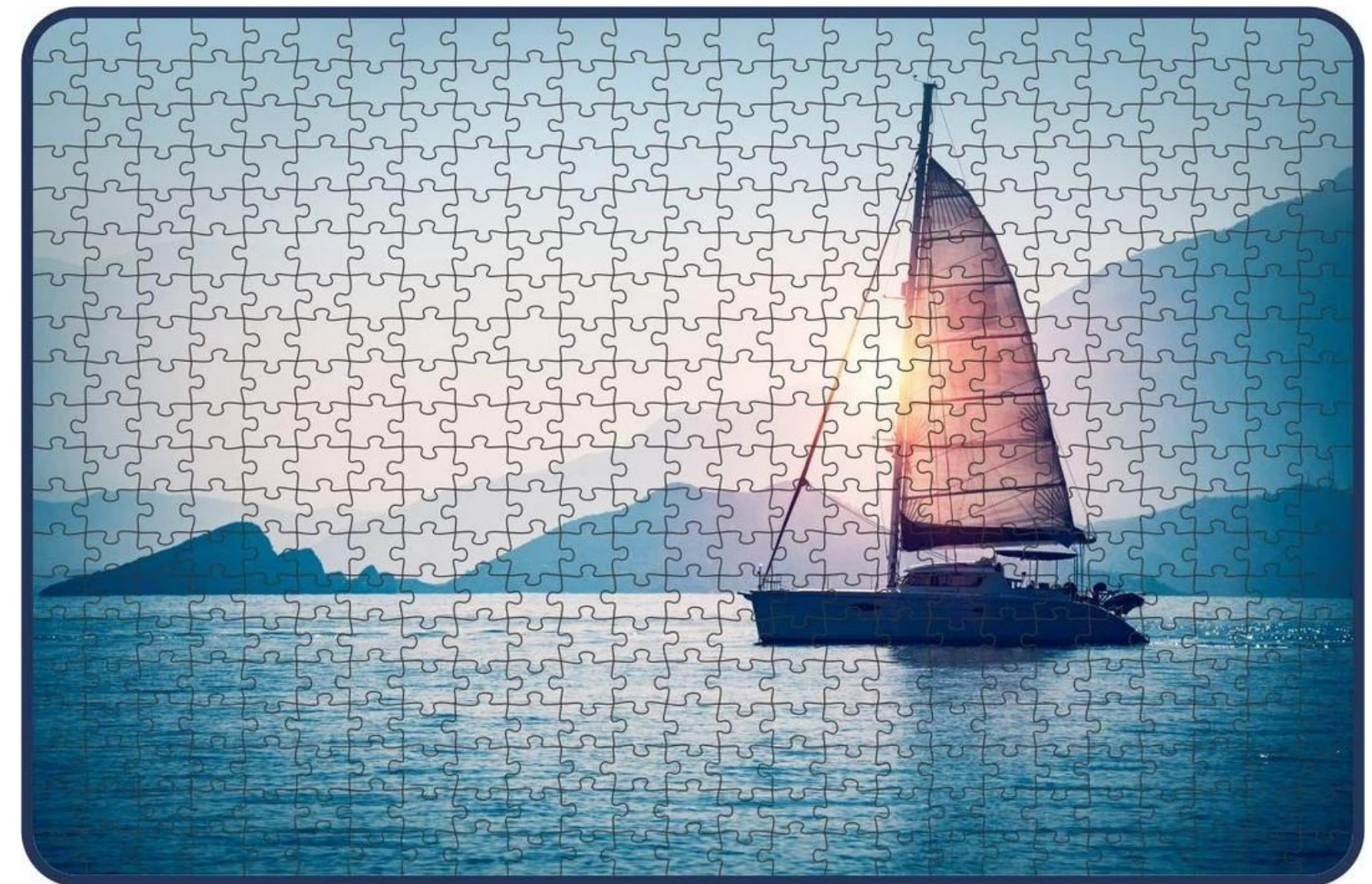
Ability to understand and produce novel combinations of familiar concepts



Let's imagine tokens are just puzzle pieces



Tokenized document



Let's imagine tokens are just puzzle pieces



Let's imagine tokens are just puzzle pieces



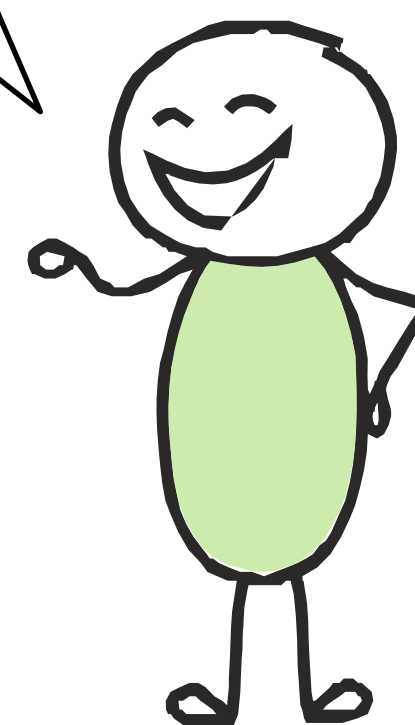
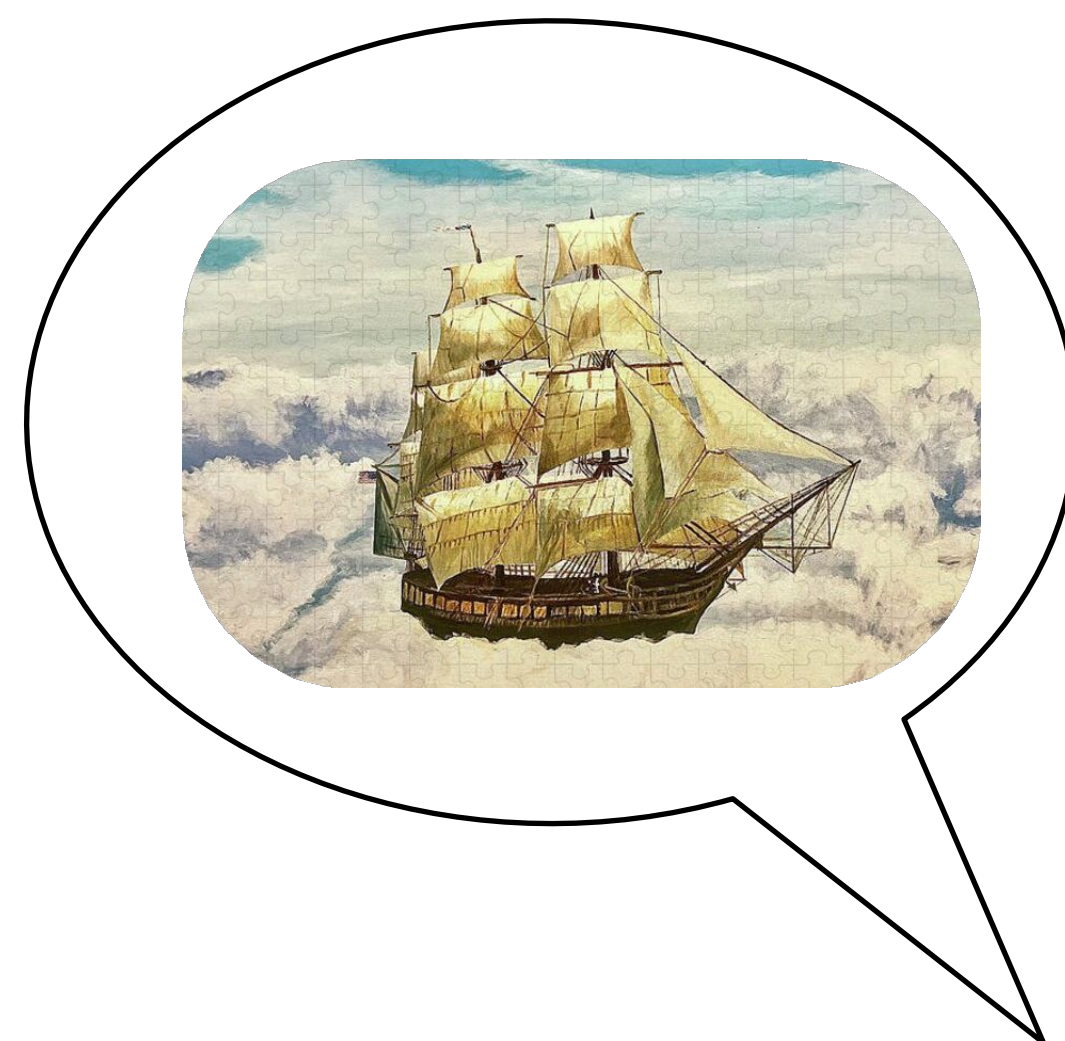
Compositional Generalization



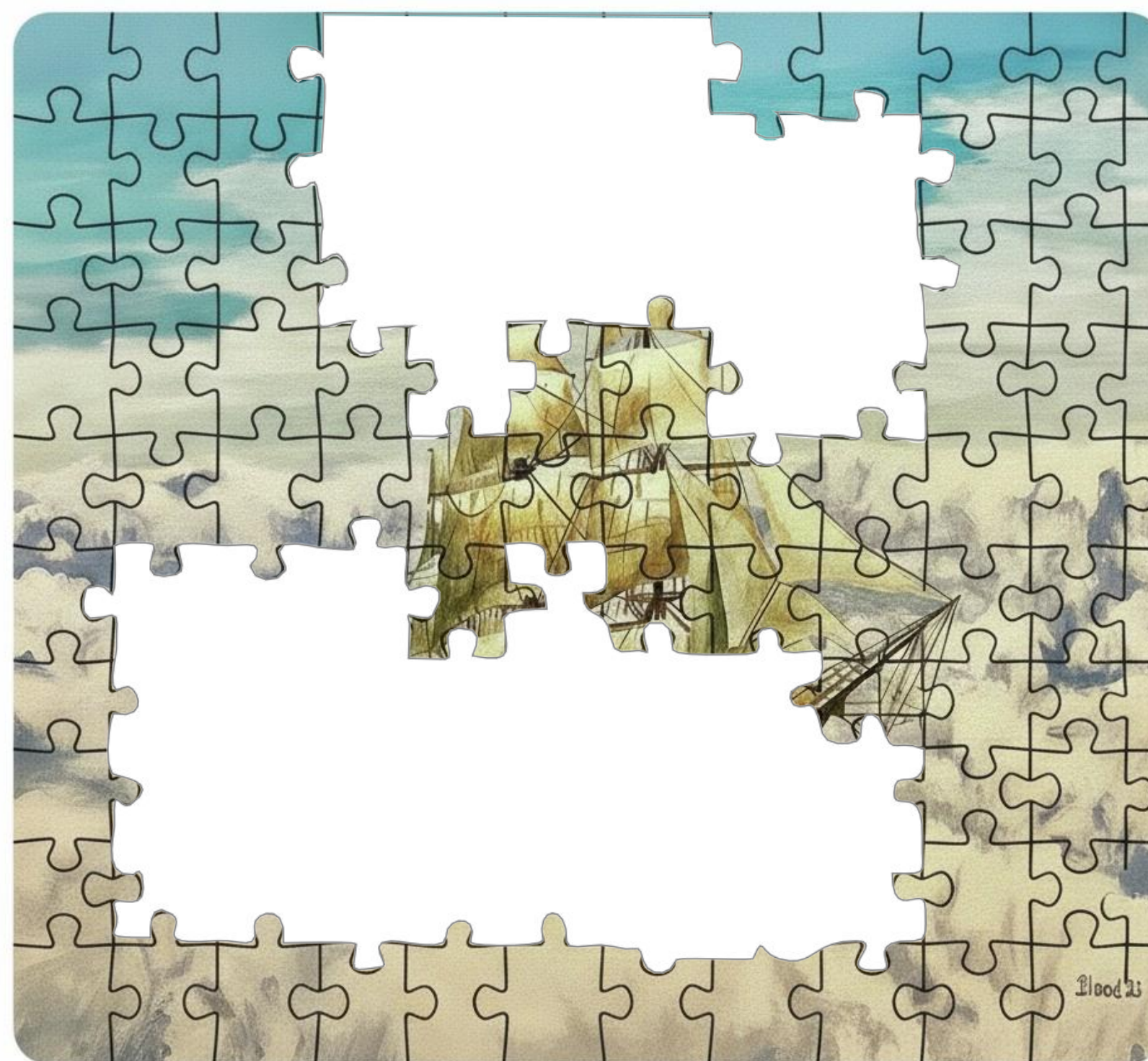
Compositional Generalization



Compositional Generalization

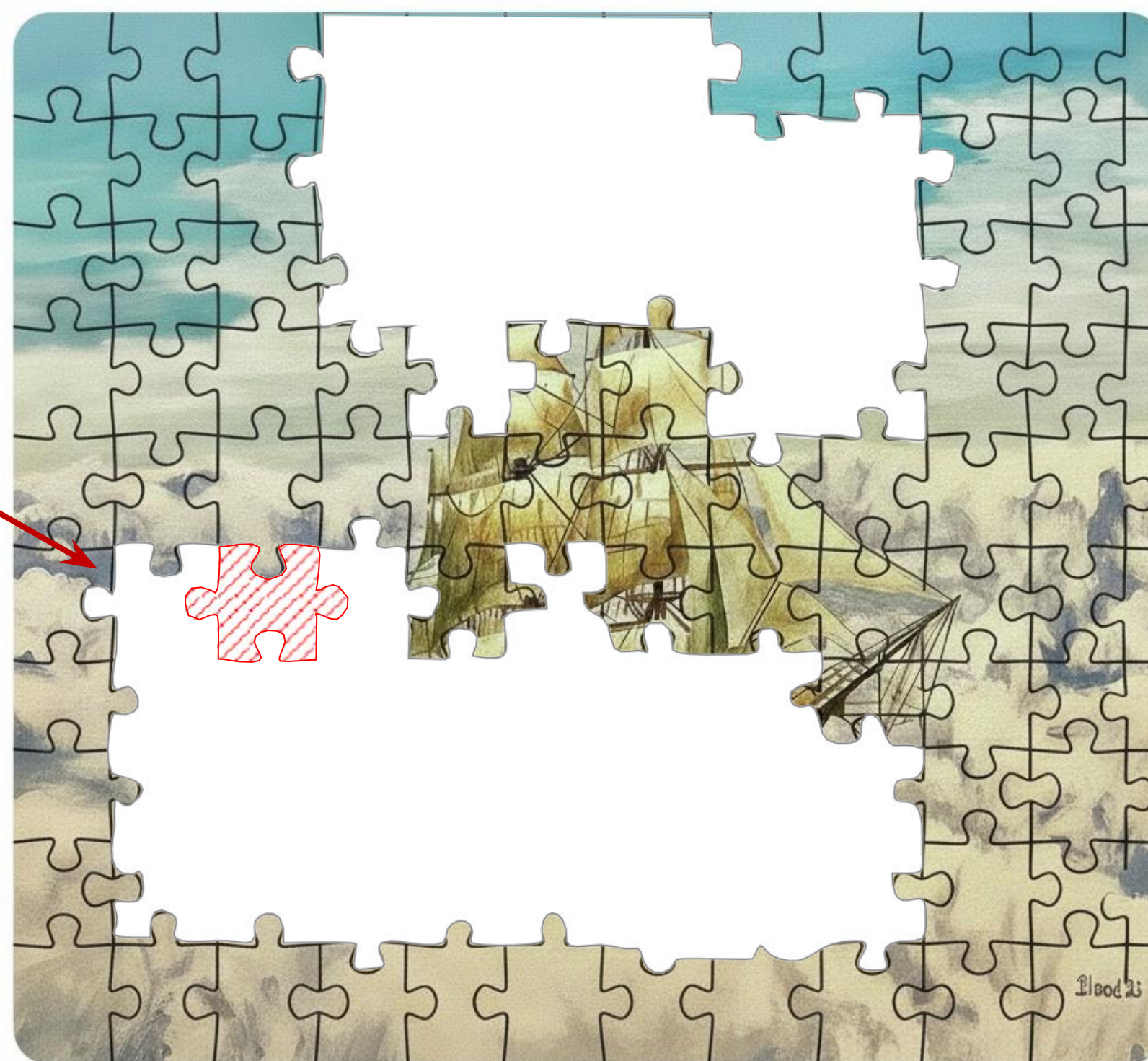


Compositional Generalization

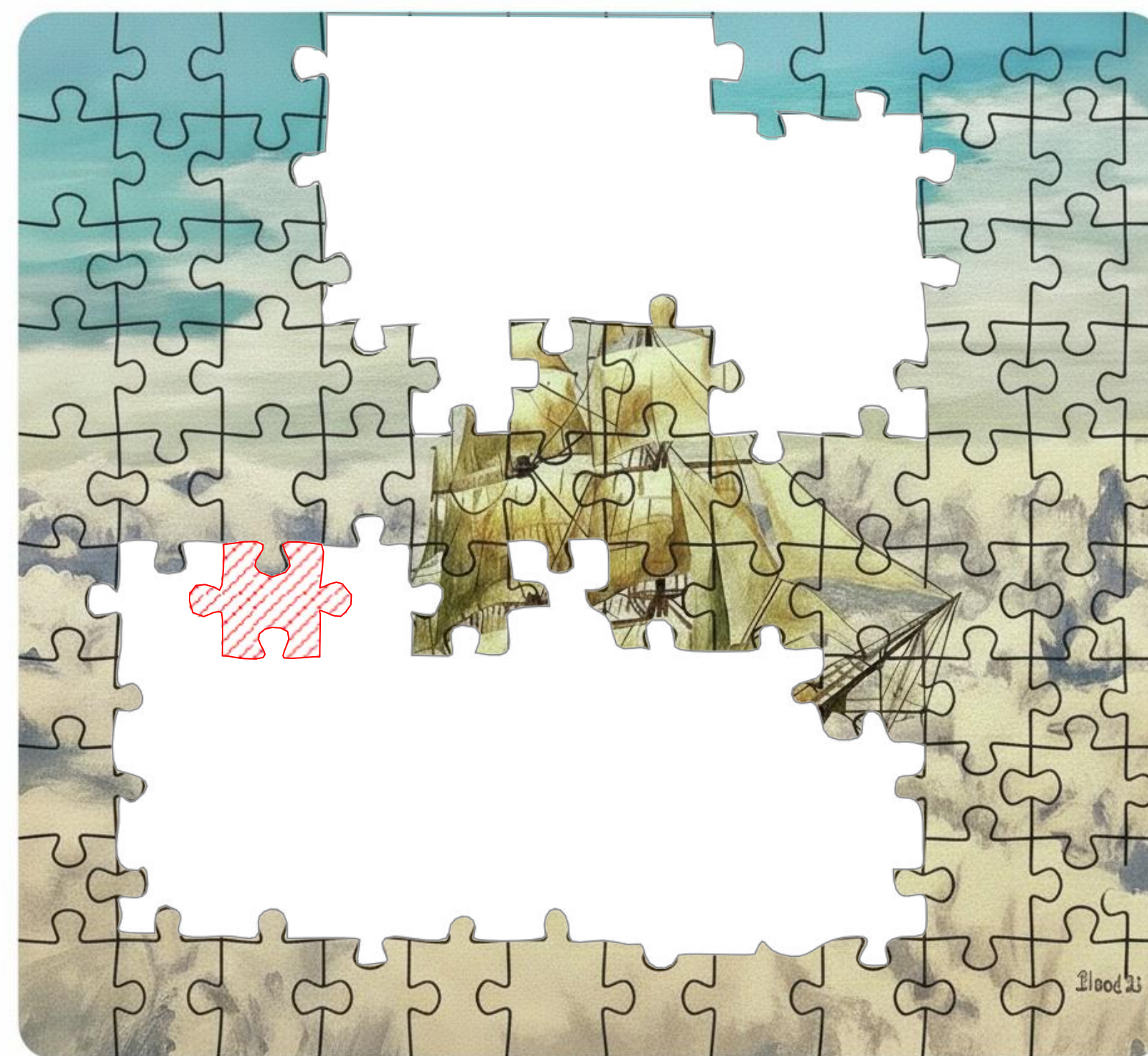


Compositional Generalization

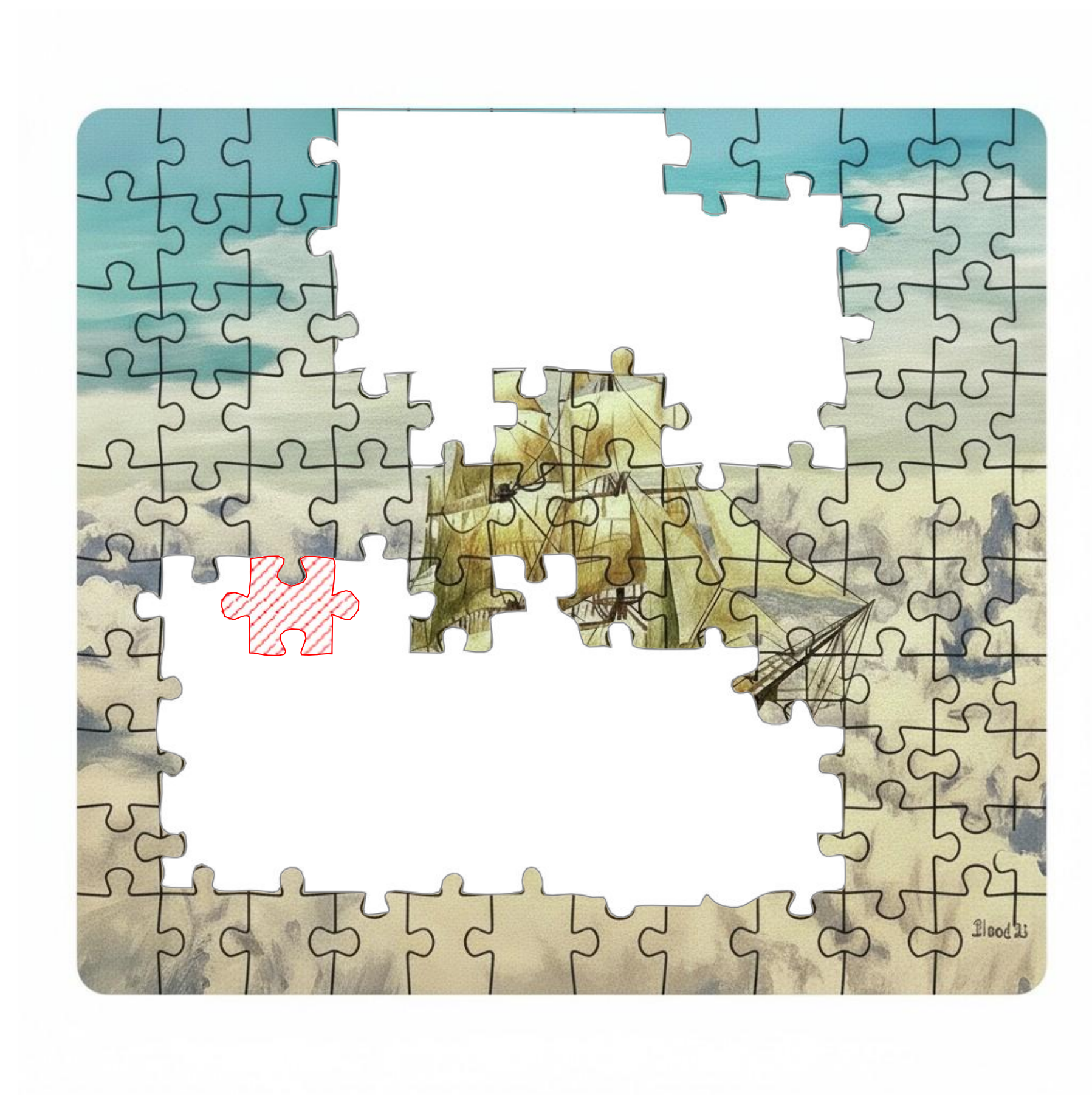
Low-probability transition



Compositional Generalization

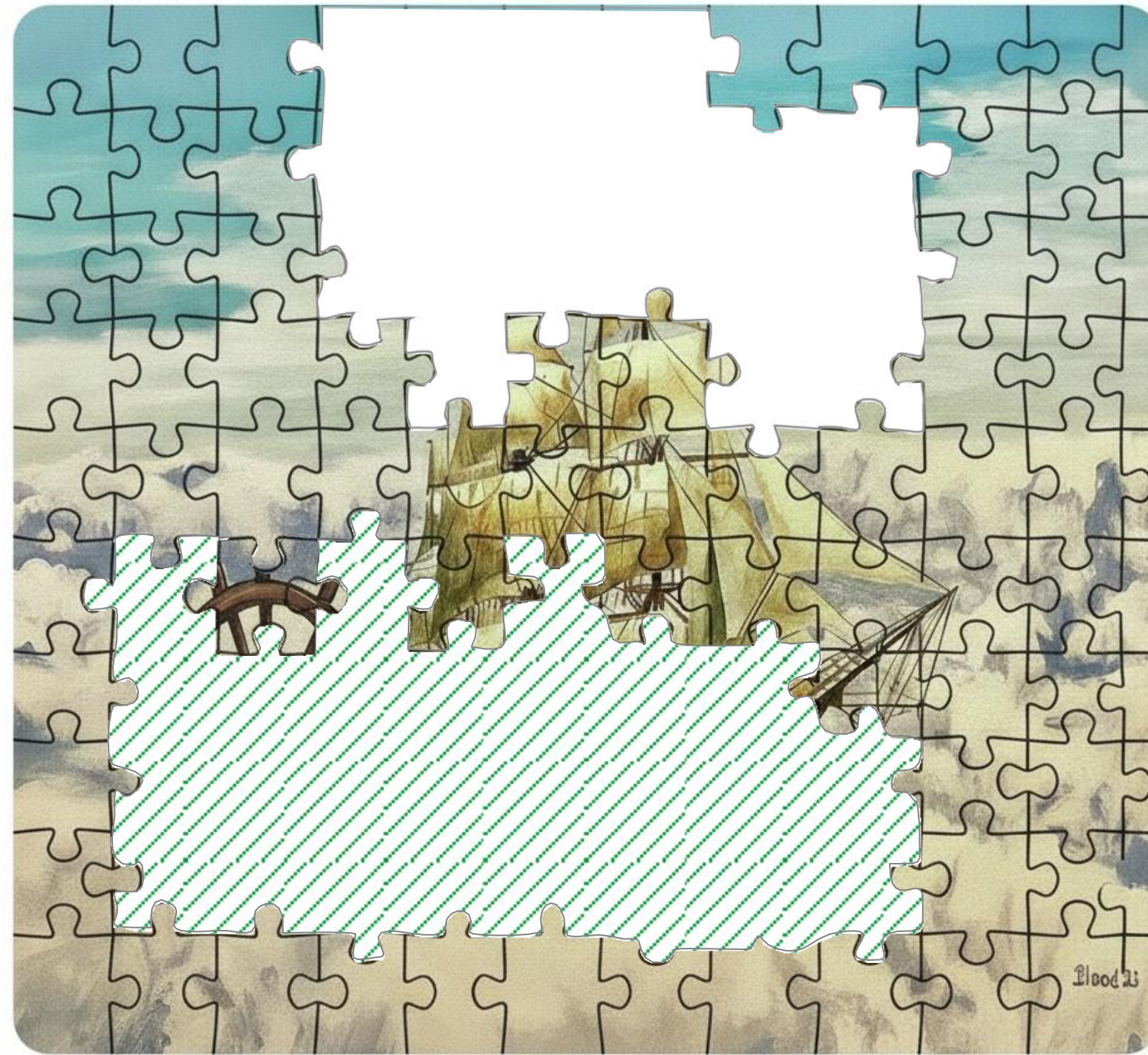


Compositional Generalization

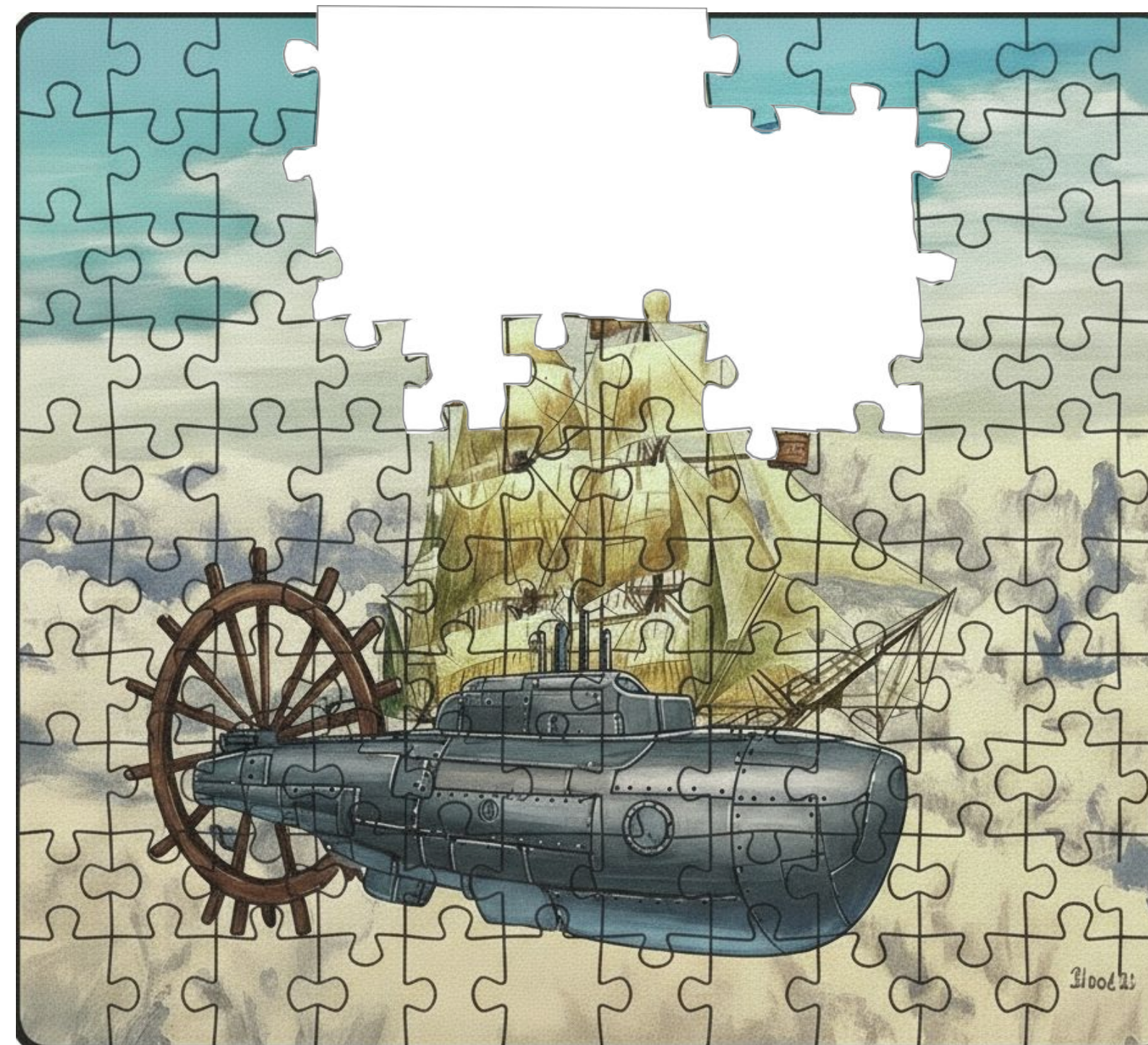


Compositional Generalization

High-probability transitions



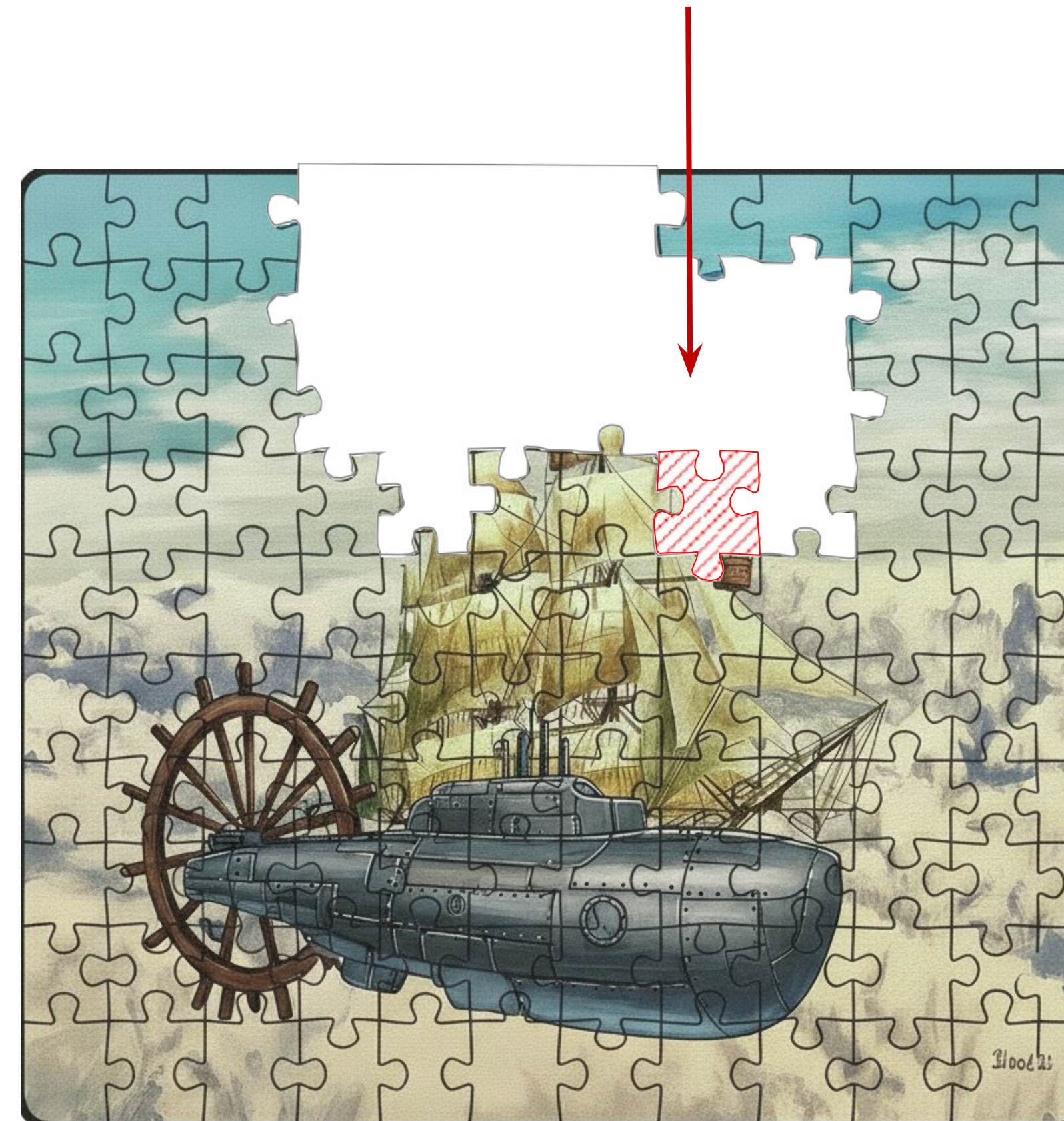
Compositional Generalization



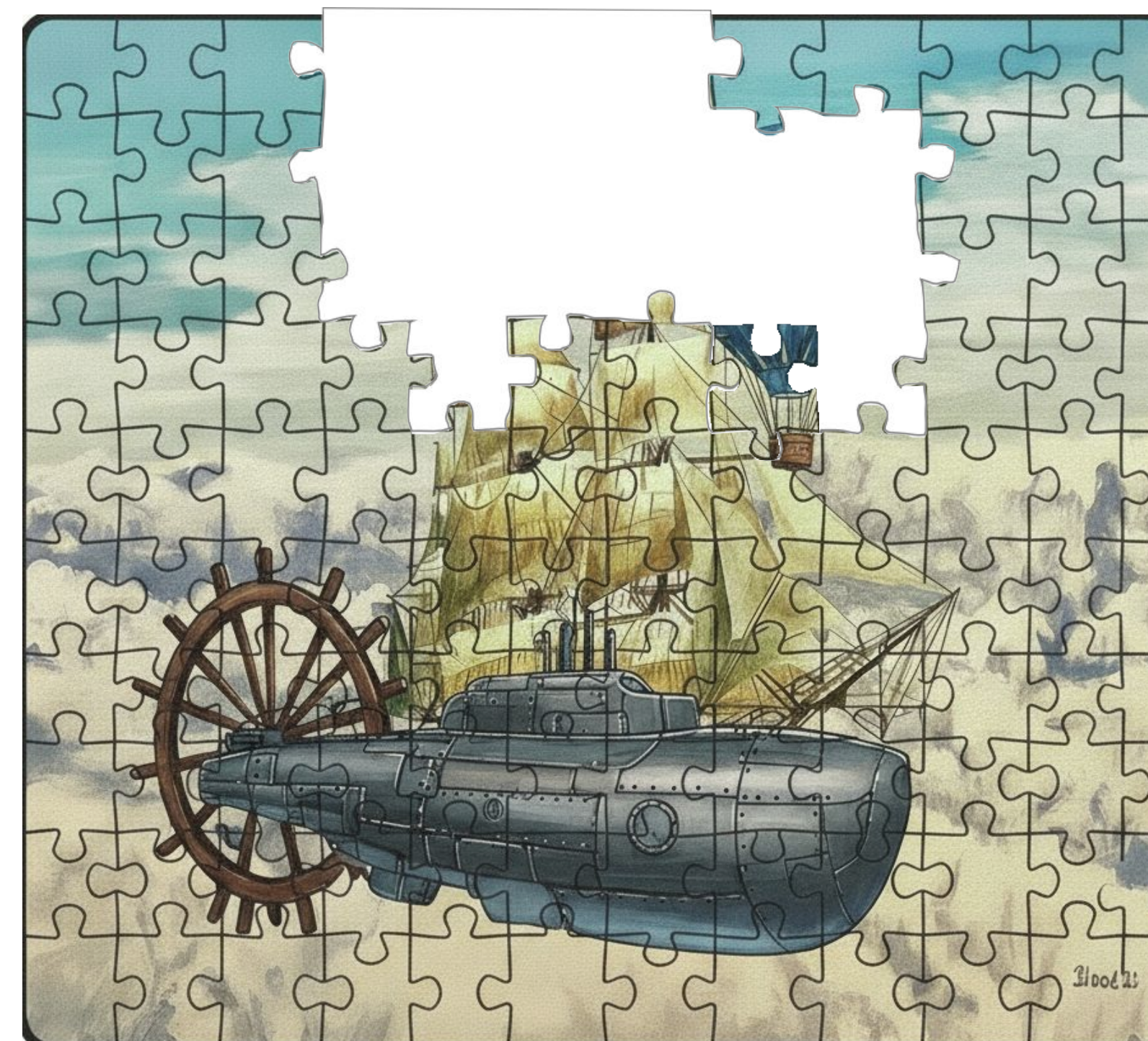
Compositional Generalization



Low-probability transition

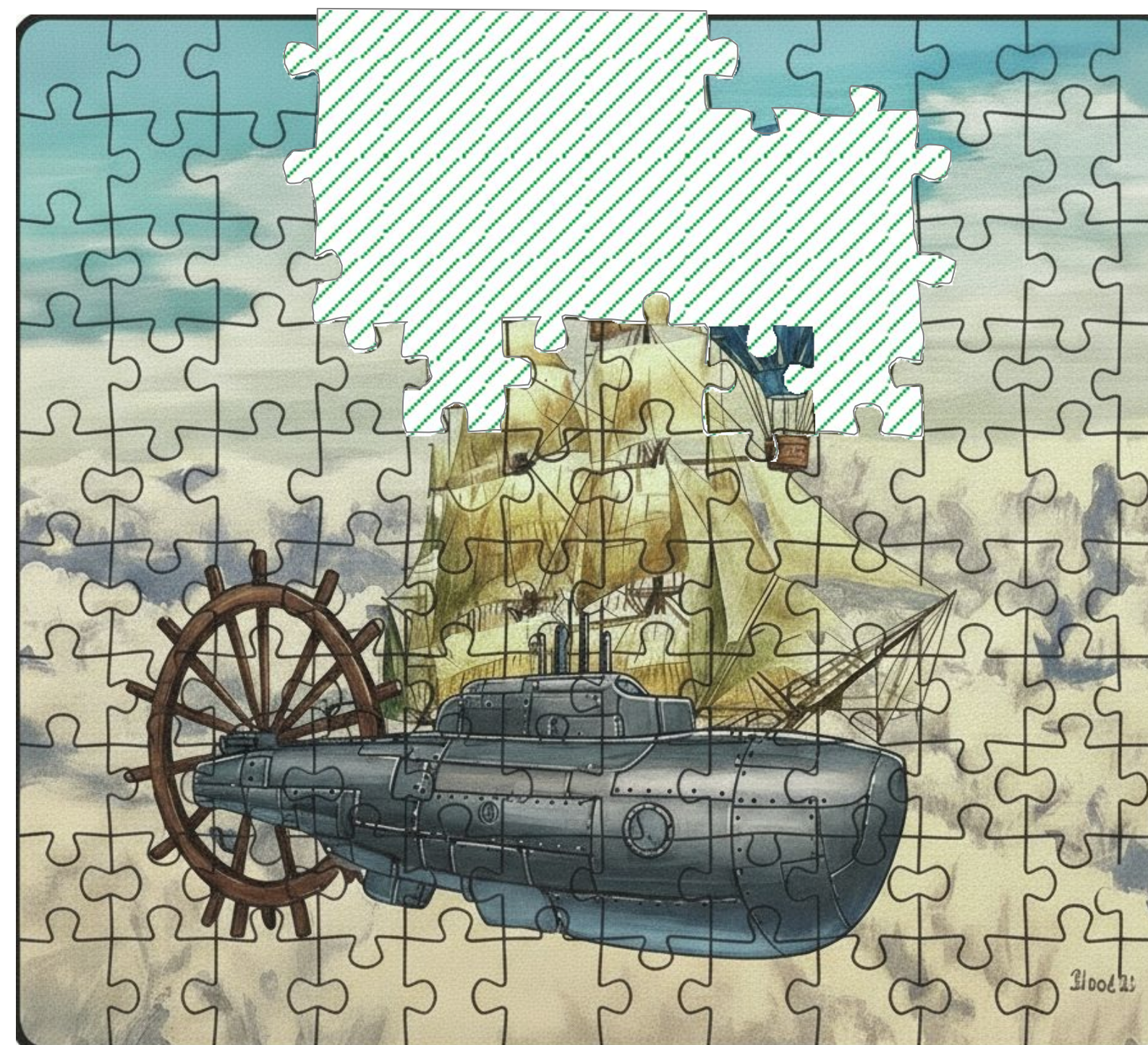


Compositional Generalization

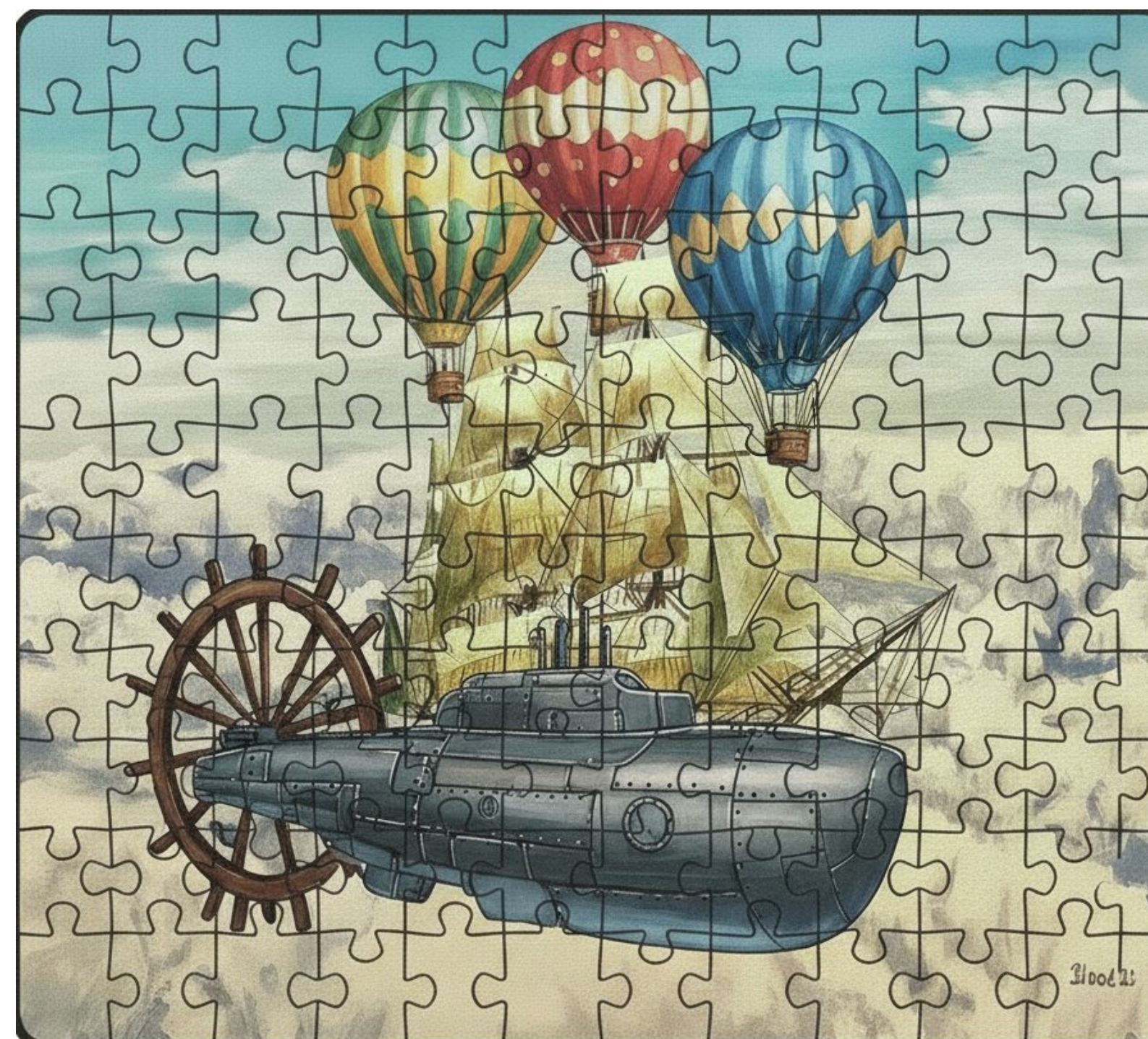


Compositional Generalization

High-probability transitions



Compositional Generalization



In-context Learning as Bayesian Inference

In-context Learning as Bayesian Inference

- (Xie et al., 2022) try to explain ICL as an implicit Bayesian inference.

Idea:

- (Pre-trained LM learn to represent “**concepts**”, i.e. the ideas described by words.
- ICL enables LMs to “**locate**” the learned concepts.

- Can formulate this intuition as a **Bayesian inference**

Prior over latent “concepts”

Likelihood describes connection between **text** and **concepts**

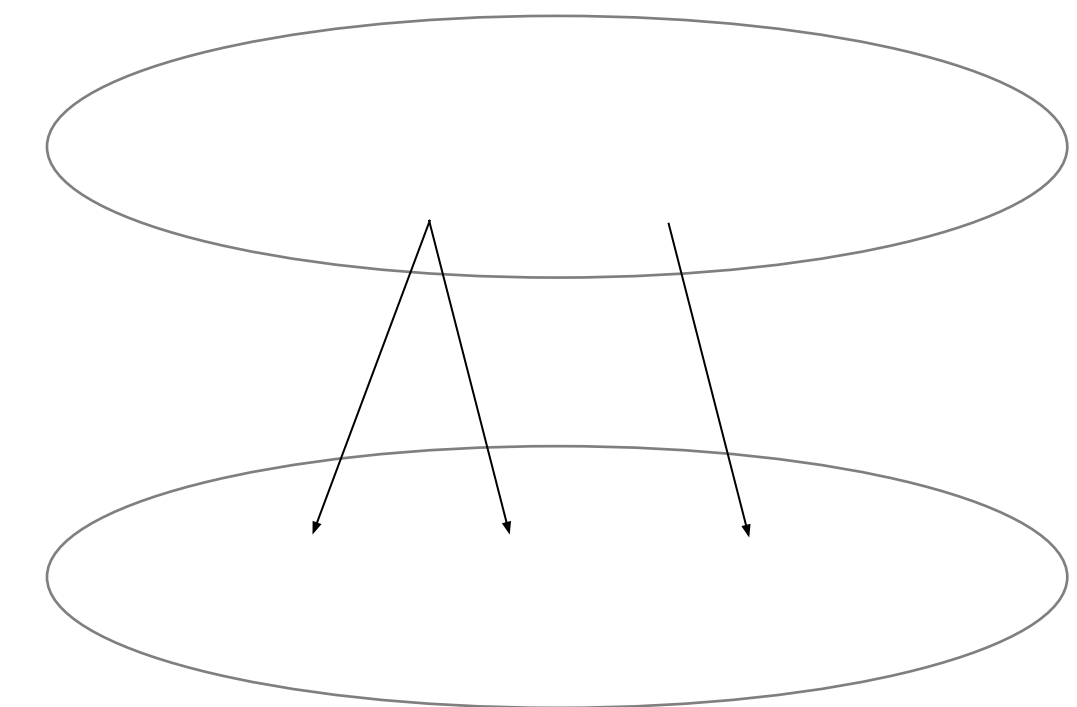
Given an incomplete doc, use Bayes formula to infer what concept is likely it is generated from and then complete the document.

- Does not explain everything.

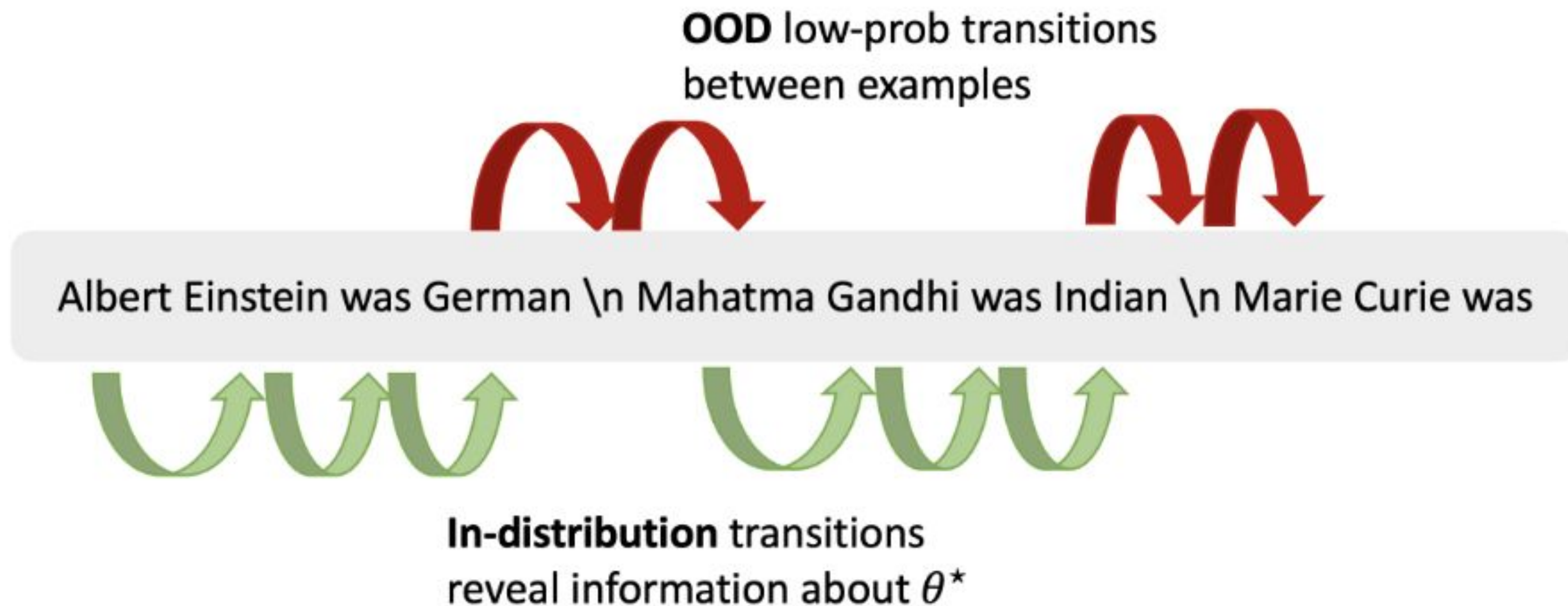
GPT-3 can handle “unseen” concepts

latent
concepts

observed documents
(e.g., demonstrations)



In-context Learning as Bayesian Inference



In-context Learning as Bayesian Inference

1. Pretraining documents are conditioned on a **latent concept** (e.g., biographical text)



Albert Einstein was a German theoretical physicist, widely acknowledged to be one of the greatest physicists of all time. Einstein is best known for developing the theory of relativity, but he also

2. Create **independent examples** from a **shared concept**. If we focus on full names, wiki bios tend to relate them to nationalities.



Input (x)	Output (y)	Delimiter
Albert Einstein was	German	\n
Mahatma Gandhi was	Indian	\n
Marie Curie was	?	...brilliant? ...Polish?

3. Concatenate examples into a **prompt** and predict next word(s). **Language model (LM)** implicitly infers the **shared concept** across examples despite the unnatural concatenation



ICL Emerges Due to Parallel Structures in Pre-training

- Natural text is abundant with “parallel structure”.

- Various forms of coreference

- Tables

- Lists

- ...

- **Hypothesis:** Such parallel structure is essential for the emergence of ICL.

Parallel Structure

For the first time in five decades, mortality rates have increased **among Palestine** refugee newborns **in Gaza**. The possible causes of this trend may include inadequate neonatal care. **We will estimate** infant and neonatal **mortality rates again in 2015** to see if this trend continues and, if so, to assess how **it can be** reversed. **Infant mortality in 2013** was 22.4 per 1000 live births compared with **20.2 in 2008** ($p \equiv 0.61$), and this change **reflected a statistically** significant

In-Context Prompt

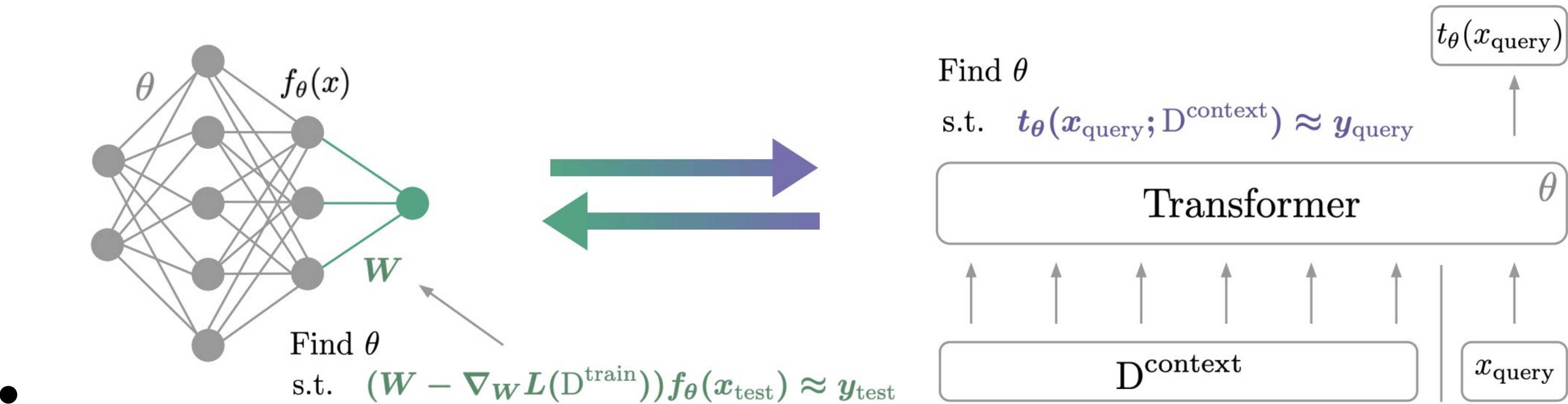
Great movie! **Sentiment: Positive**. I hate the movie! **Sentiment: Negative**. This movie is awesome. Sentiment: Positive.

In-context Learning as Gradient Descent?

In-context Learning as Gradient Descent?

[von Oswald et al. 2022; Akyurek et al. 2022; Dai et al. 2022, ...]

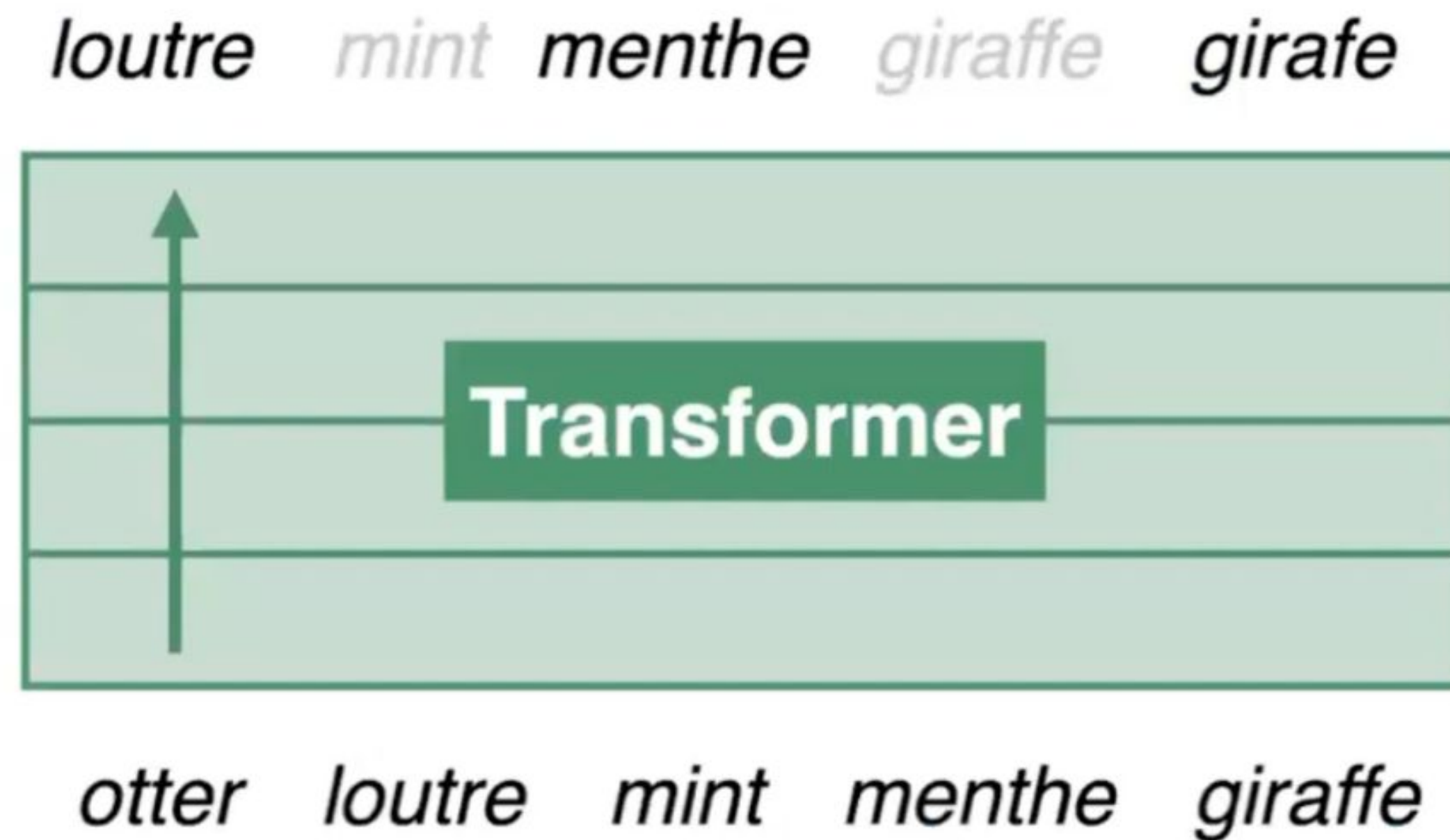
- Hypothesis: ICL is implicitly equivalent to SGD on in-context demonstrations



[Do Pre-trained Transformers Really Learn In-context by Gradient Descent?, 2024]

ICL for translation

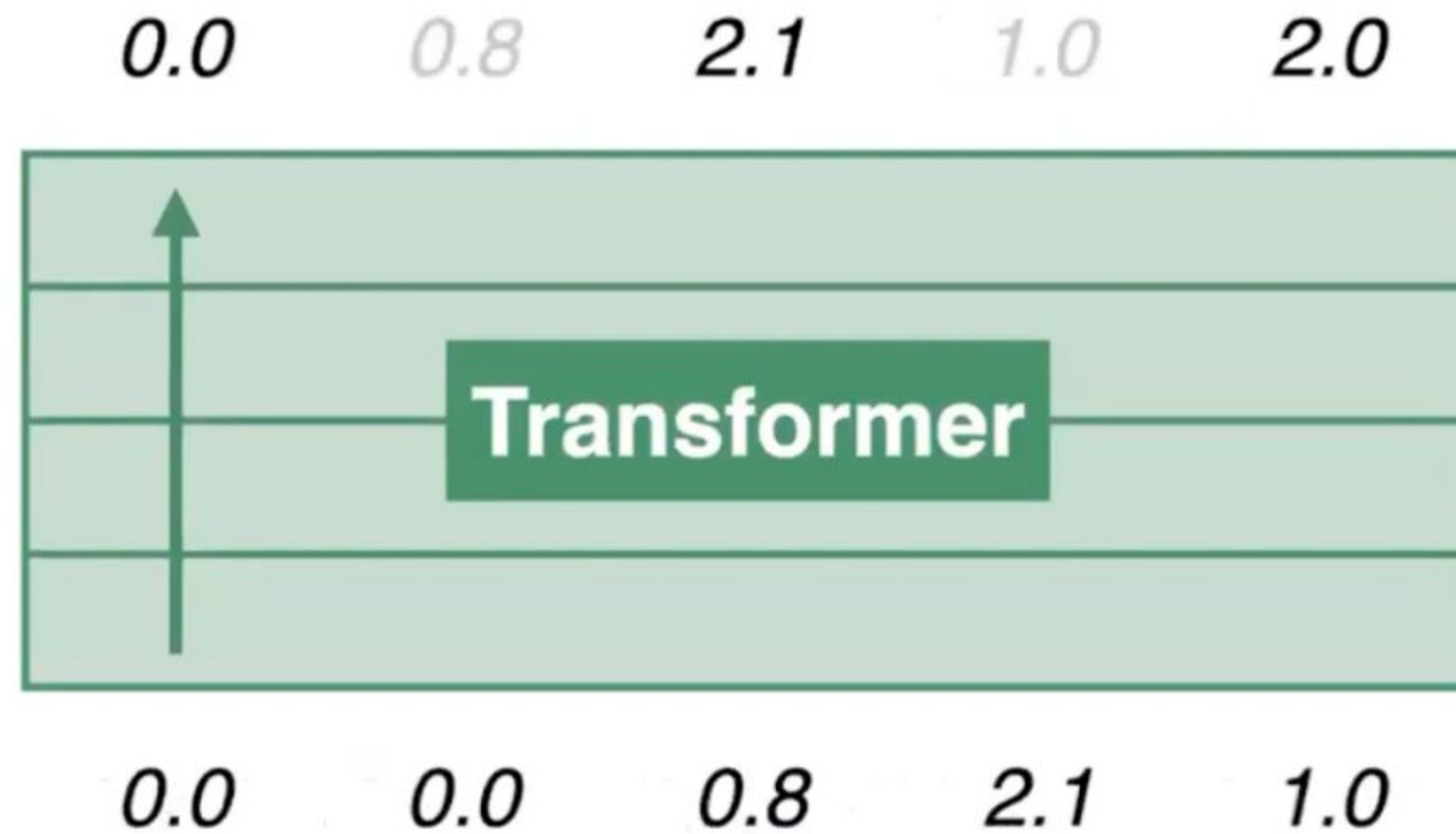
x	y
<i>otter</i>	<i>loutre</i>
<i>mint</i>	<i>menthe</i>
<i>giraffe</i>	???



ICL for regression

x	y
0.0	0.0
0.8	2.1
1.0	???

$$y = w^T x$$



OLS regression

x	y
0.0	0.0
0.8	2.1
1.0	???

$$y = w^T x$$

0.0 0.8 2.1 1.0 2.0

$$\hat{w} = (X^T X)^{-1} X^T f(x)$$

$$\hat{f}(x) = \hat{w}^T x_n$$

0.0 0.0 0.8 2.1 1.0

Transformer layer and OLS

The Transformer should solve: $f_t(x_{1:t}) = \hat{W}_t x_t = X_t X_{t-1}^T (X_{t-1} X_{t-1}^T + \lambda I)^{-1} x_t$

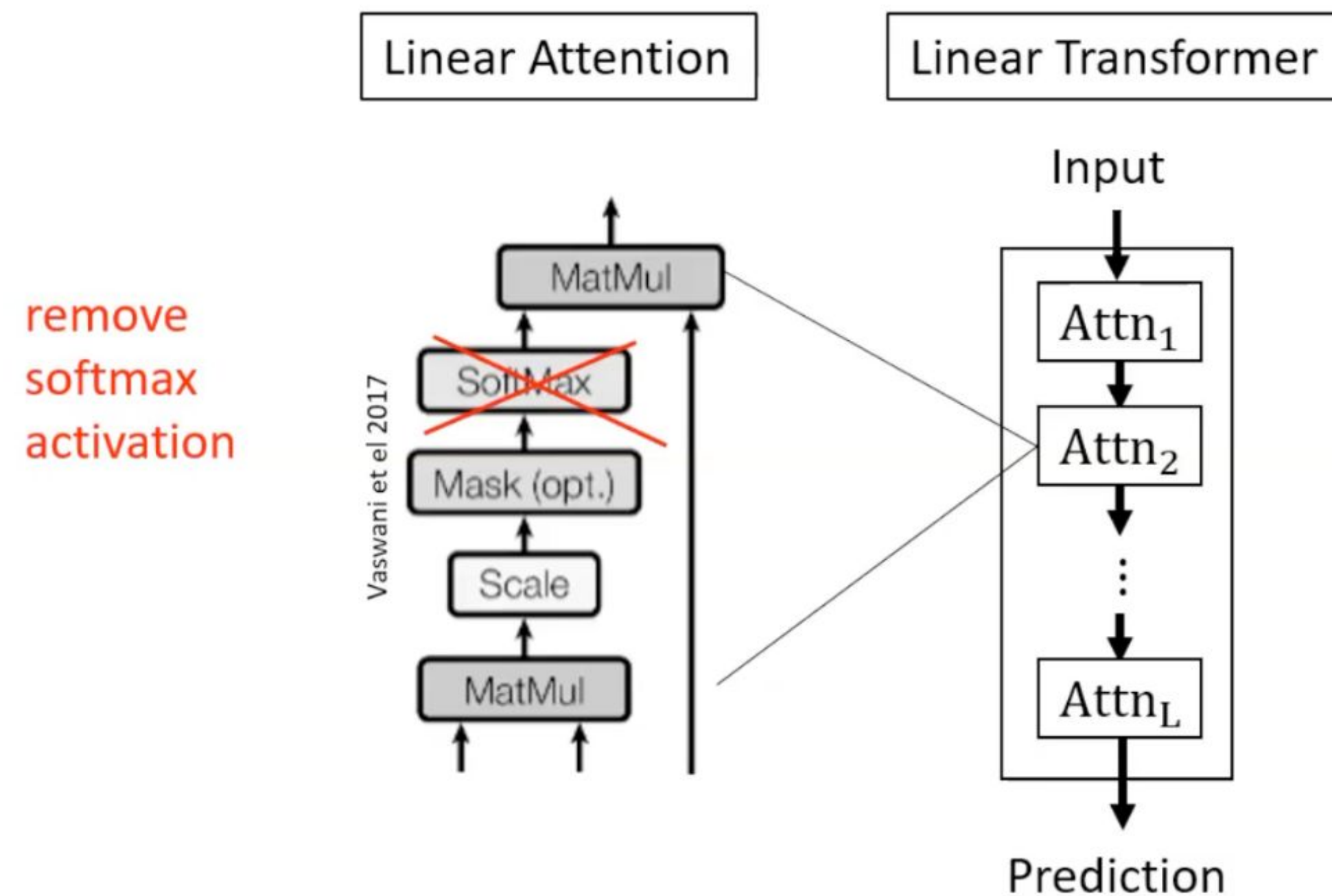
We provide **weight constructions** for the following (after copying):

- 1) One causal *linear* self-attention **can implement a step of GD for t internal optimization problems** i.e. $f_t(x_{1:t}) = \hat{W}_t x_t = \eta X_t X_{t-1}^T x_t$ in parallel
- 2) One causal linear self-attention can compute the next summand of **t projected truncated Neumann series in parallel**

$$(X_{t-1} X_{t-1}^T + \lambda I)^{-1} x_t = \sum_{k=0}^{\infty} ((1 - \lambda)I - X_{t-1} X_{t-1}^T)^k x_t$$

This implies that in **$\log(\log(1/\epsilon))$ -layers** the Transformer can approximate **t** matrix inverses to epsilon error.

How does linear transformer in ICL?



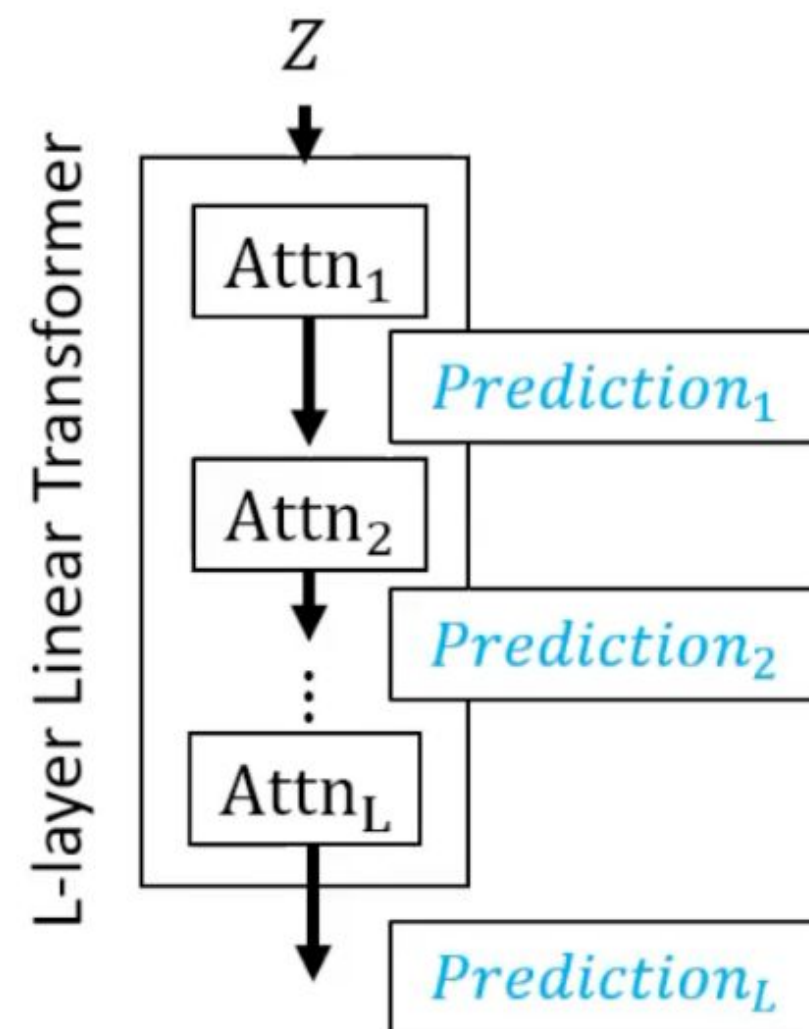
- For a **single layer linear** transformer trained on the random instances of linear regression (esp. when the in-context samples are non-isotropic), the set of global minimizers corresponds to implementing one step of a (preconditioned) gradient descent on the least square objective. Here the least square objective is based on the examples given in the prompt.
- For a **L-layer linear transformer** trained on the random instances of linear regression, a subset of stationary points corresponds to implementing L steps of a (preconditioned) gradient descent on the least square objective.

Transformer learns to implement gradient descent

Input: $Z = \begin{bmatrix} x^{(1)} & x^{(2)} & \cdots & x^{(n)} & x^{(n+1)} \\ y^{(1)} & y^{(2)} & \cdots & y^{(n)} & 0 \end{bmatrix} \in \mathbb{R}^{(d+1) \times (n+1)}$

demonstrations query

Goal: **predict** $y^{(n+1)}$



$Prediction_k$: The Transformer's prediction for $y^{(n+1)}$ at layer k , given input Z

Training objective:

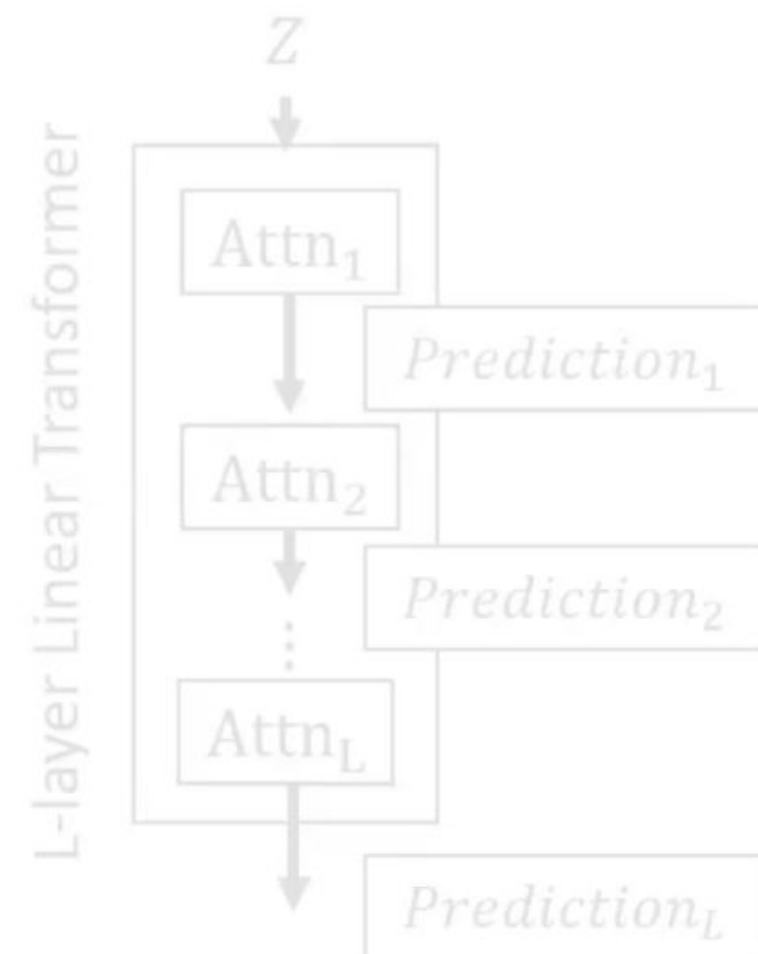
$$\mathbb{E}_{(Z, \theta^*)} \left[\left(Prediction_L - \underset{\text{(True label)}}{y^{(n+1)}} \right)^2 \right]$$

Transformer learns to implement gradient descent

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demonstrations
query

Goal: predict $y^{(n+1)}$

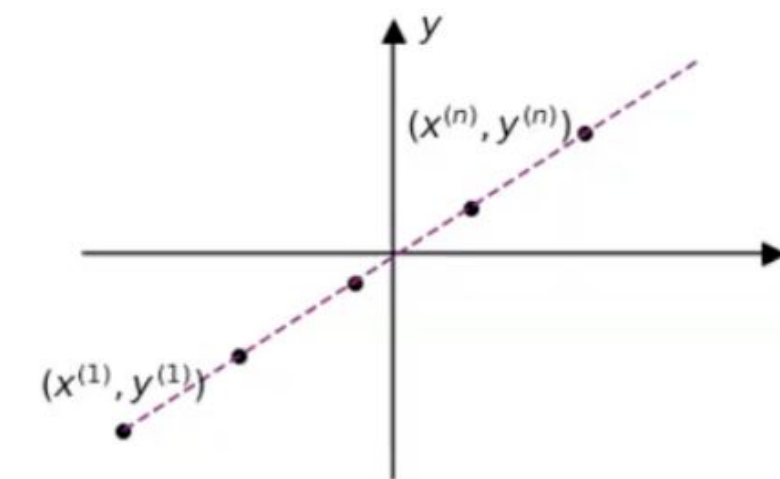


Training objective:

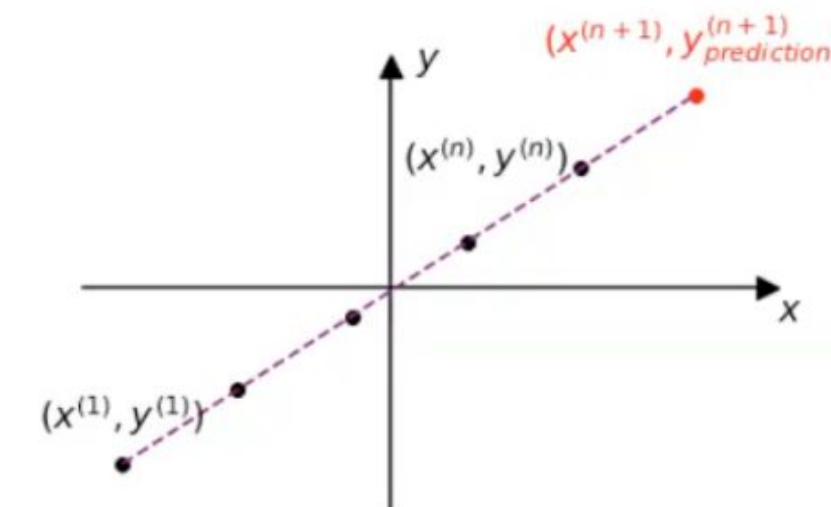
$$\mathbb{E} \left[\left(\underset{\text{(True label)}}{Prediction_L} - y^{(n+1)} \right)^2 \right]$$

Given n demonstrations, how do we predict $y^{(n+1)}$?

Step 1: Find best-fit $\hat{\theta}$ for demonstrations.



Step 2: Use $\hat{\theta}$ to predict $y^{(n+1)}$

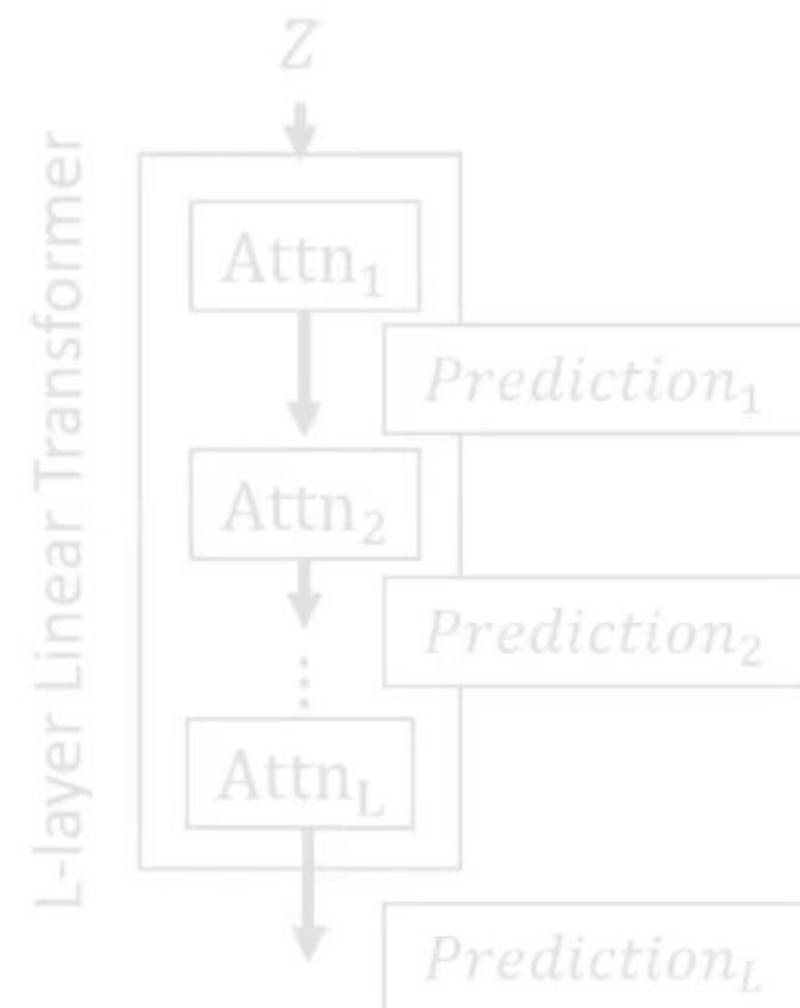


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demonstrations
query

Goal: predict $y^{(n+1)}$

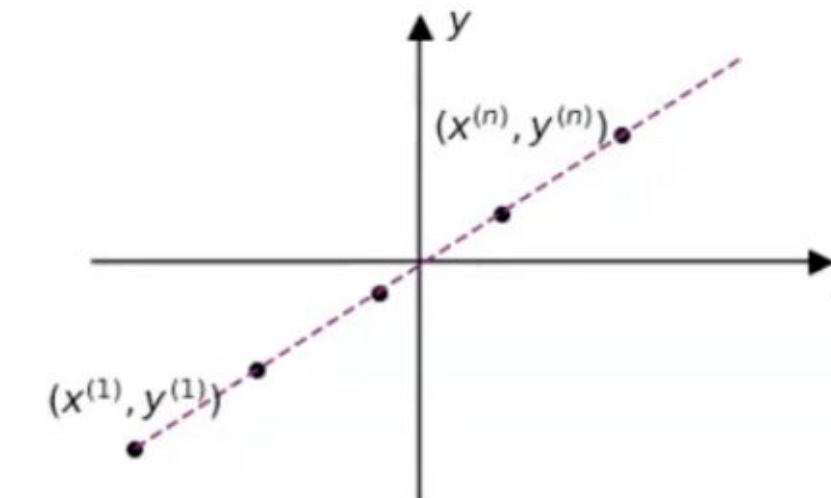


Training objective:

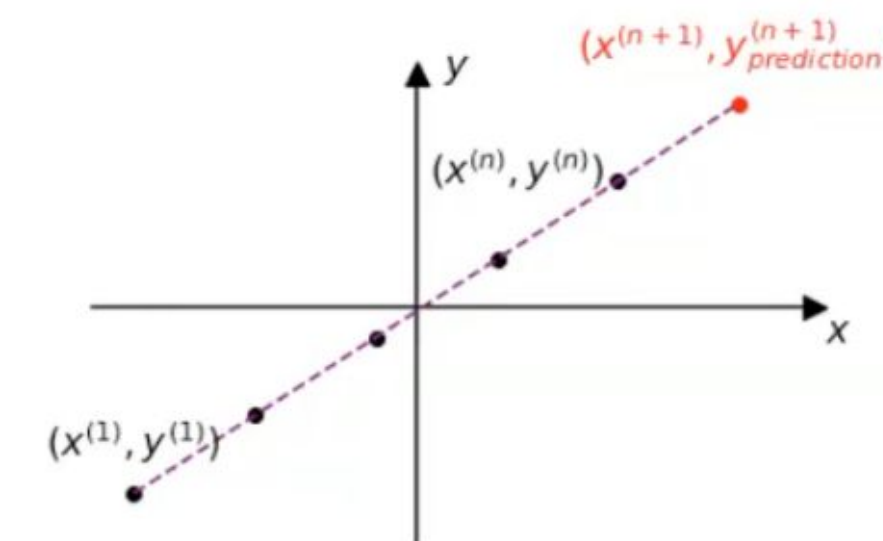
$$\mathbb{E} \left[\left(\underset{\text{(True label)}}{\text{Prediction}_L} - y^{(n+1)} \right)^2 \right]$$

Given n demonstrations, how do we predict $y^{(n+1)}$?

Step 1: Find best-fit $\hat{\theta}$ for demonstrations.



Step 2: Use $\hat{\theta}$ to predict $y^{(n+1)}$



$\hat{\theta}$ minimizes the Empirical Loss

$$R(\theta) = \sum_{i=1}^n (y^{(i)} - \theta^\top x^{(i)})^2$$

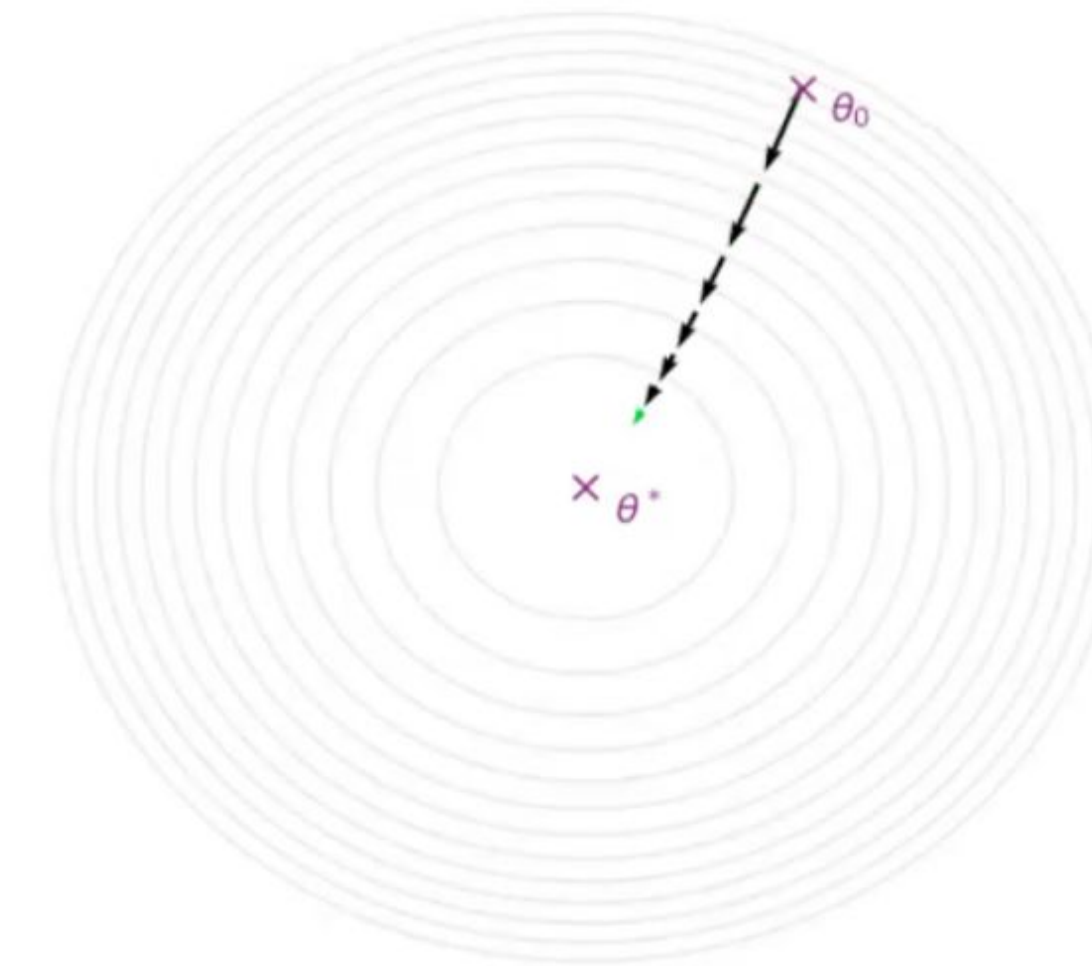
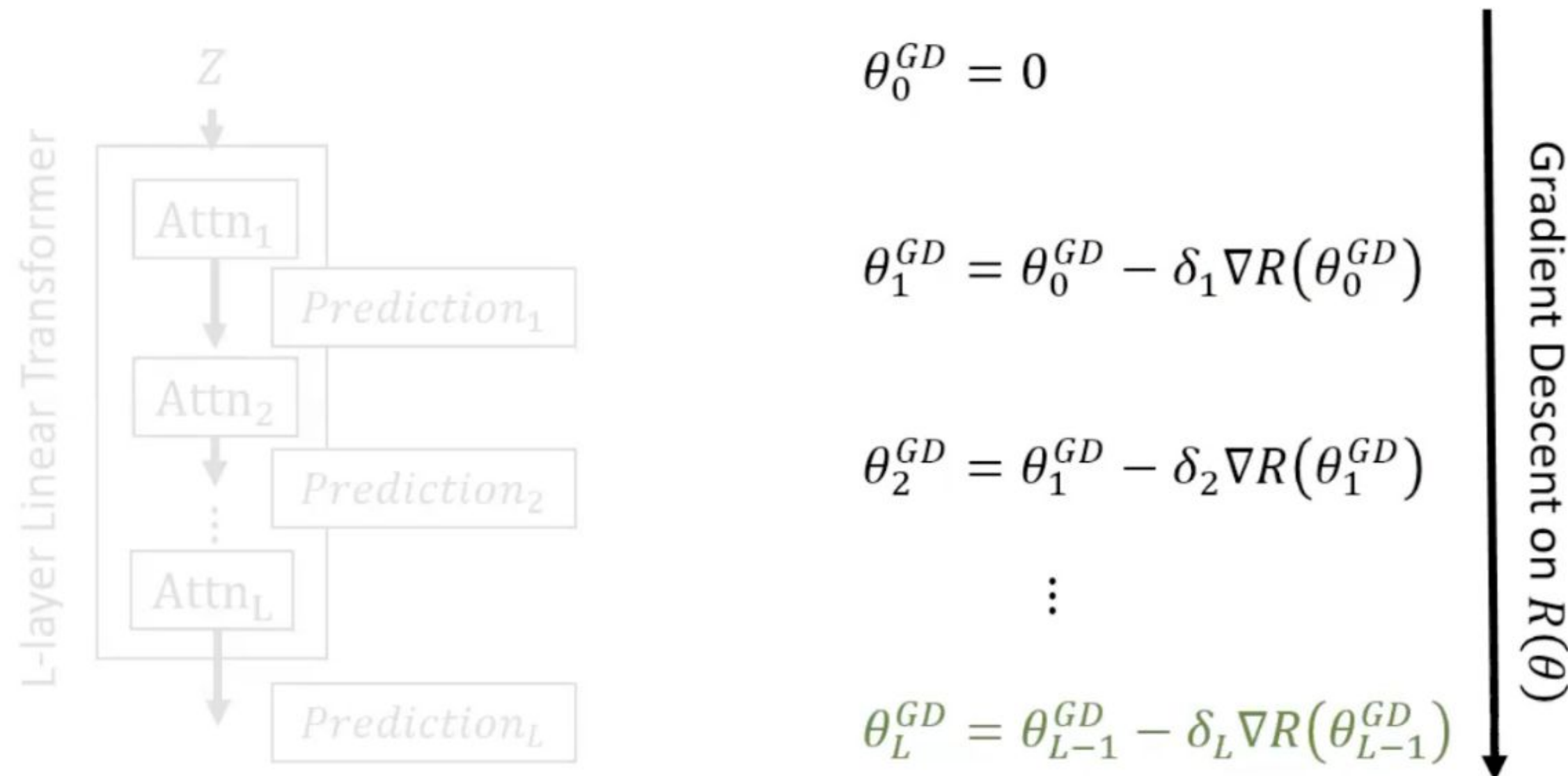
sum over n
demonstrations

Transformer learns to implement gradient descent

Input: $Z = \begin{bmatrix} x^{(1)} & x^{(2)} & \dots & x^{(n)} & x^{(n+1)} \\ y^{(1)} & y^{(2)} & \dots & y^{(n)} & 0 \end{bmatrix} \in \mathbb{R}^{(d+1) \times (n+1)}$

demonstrations
query

Goal: predict $y^{(n+1)}$



Training objective:

$$\mathbb{E} \left[\left(\text{Prediction}_L - \underset{\text{(True label)}}{y^{(n+1)}} \right)^2 \right]$$

$\hat{\theta}$ minimizes the Empirical Loss

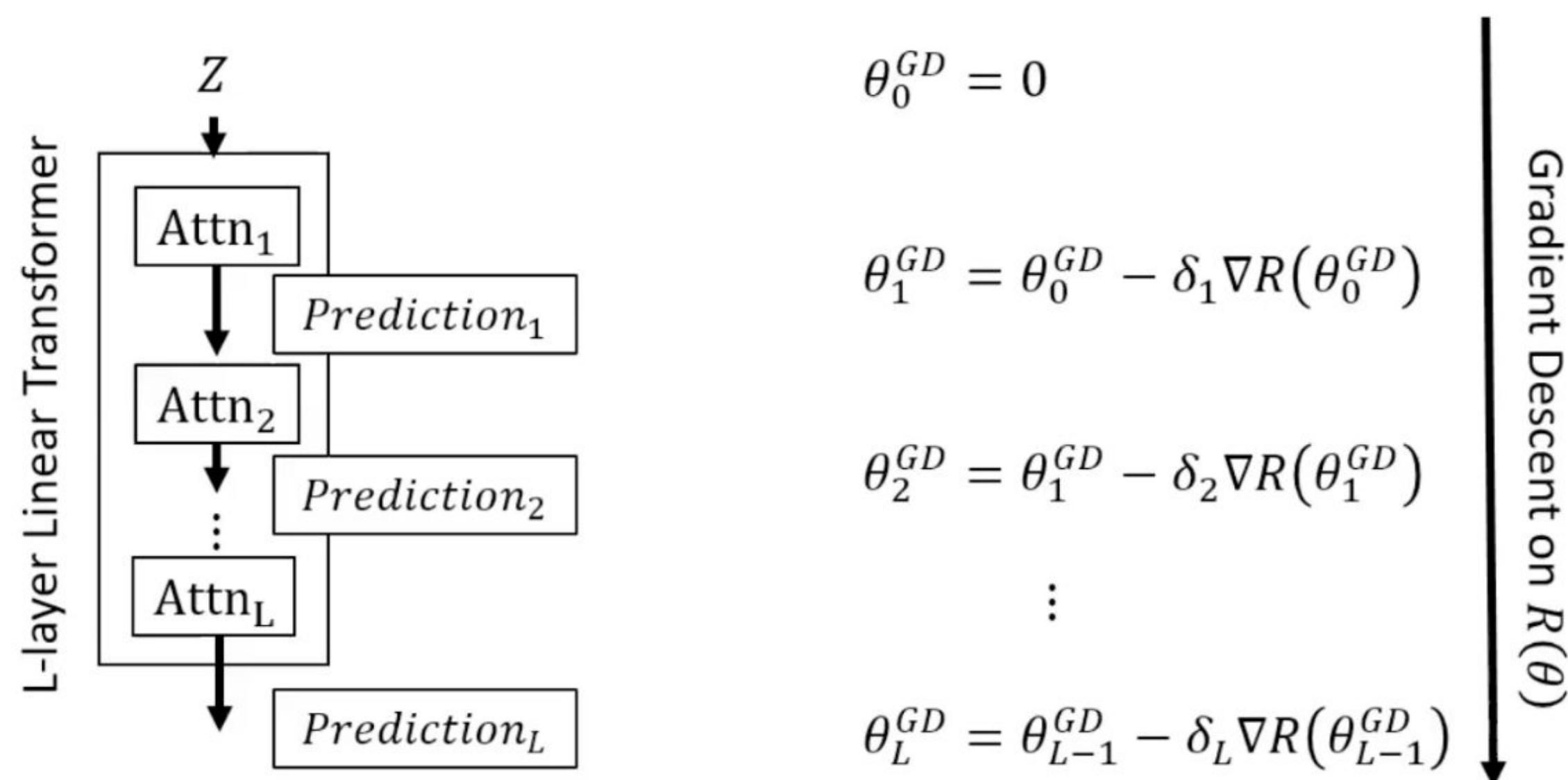
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Transformer learns to implement gradient descent

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demonstrations
query

Goal: predict $y^{(n+1)}$



Theorem 3 [ACDS 23]

Assume $x^{(i)} \sim \mathcal{N}(0, I)$, $\theta^* \sim \mathcal{N}(0, I)$. There **exist stationary points** of the training objective, such that the Transformer prediction at layer k is the same as k steps of gradient descent on $R(\theta)$, i.e.

$$\text{Prediction}_k = \langle \theta_k^{GD}, x^{(n+1)} \rangle,$$

for $k = 1 \dots L$

Training objective:

$$\mathbb{E} \left[\left(\text{Prediction}_L - y^{(n+1)} \right)^2 \right]$$

(True label)

$\hat{\theta}$ minimizes the Empirical Loss

$$R(\theta) = \sum_{i=1}^n (y^{(i)} - \theta^\top x^{(i)})^2$$

Theorem 1 [ACDS 23]

If layer = 1, the **global minimum** of the training objective coincides with one step of gradient descent.

Brief history

- + : Formalize the in-context learning problem
- + : Experiments: Transformers can be trained to achieve good ICL loss
- : Consider simple functions

Language Models are Few Shot Learners
Brown, Mann, Ryder, Subbiah, et al 2020

What Can Transformers Learn In-Context? A Case Study of Simple Function Classes
Garg, Tsipras, Liang, Valiant 2022

- + : Propose mechanism for ICL via OLS
- + : Experimental evidence by comparing loss over different tasks
- : Complicated construction, difficult to learn in practice
- : Consider simple functions

What learning algorithm is in-context learning? Investigations with linear models
Akyurek, Schuurmans, Andreas, Ma, Zhou 2023
Transformers as statisticians: Provable in-context learning with in-context algorithm selection
Lin, Bai, Mei 2023

- + : Detailed mechanism – ICL via OLS via Gradient Descent
- + : Some experimental evidence that proposed parameters are being learned
- : Linear Transformer applied to learning linear functions

Linear Transformers Are Secretly Fast Weight Programmers
Schlag, Irie, Schmidhuber 2021
Transformers Learn In-Context by Gradient Descent
Von Oswald, Niklasson, Randazzo, Sacramento, Mordvintsev, Zhmoginov, Vladymyrov 2023

- + : Theoretical proof that Transformer implement GD at local/global minimum
- + : Experimental evidence that proposed parameters are exactly being learned
- : Linear Transformer applied to learning linear functions

Transformers learn to implement preconditioned gradient descent for in-context learning
Ahn, Cheng, Daneshmand, Sra 2023
Trained transformers learn linear models in-context
R Zhang, S Frei, PL Bartlett
One step of gradient descent is provably the optimal in-context learner with one layer of linear self-attention
Mahankali, Hashimoto, Ma 2023

- + : Nonlinear Transformer applied to Nonlinear Functions
- + : ICL via functional gradient descent, related to choice of nonlinear activation
- : Several restrictions apply

Transformers implement functional gradient descent to learn non-linear functions in context
Cheng, Chen, Sra, 2024

Why Do We Care About In-Context Learning?

Practically Useful

Intellectually Intriguing

In-Context Learning: Practically Useful

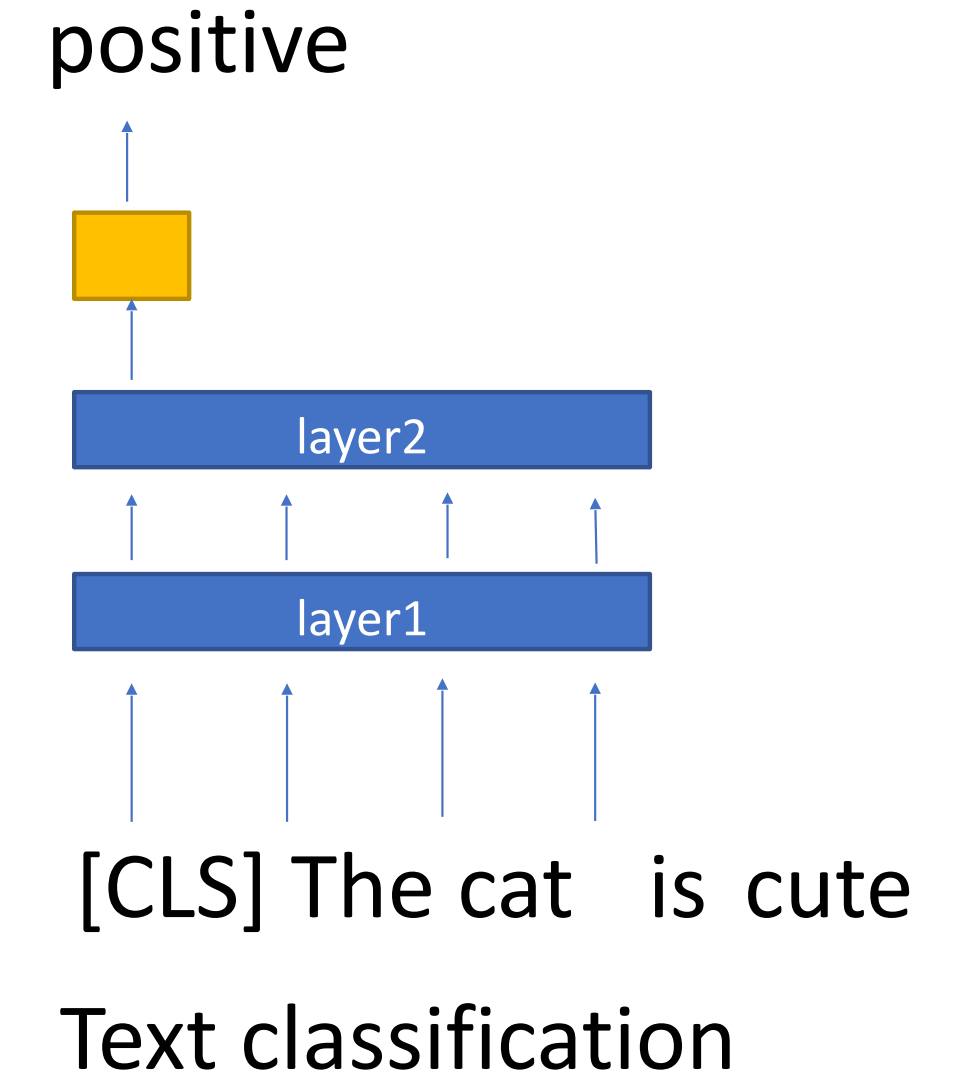
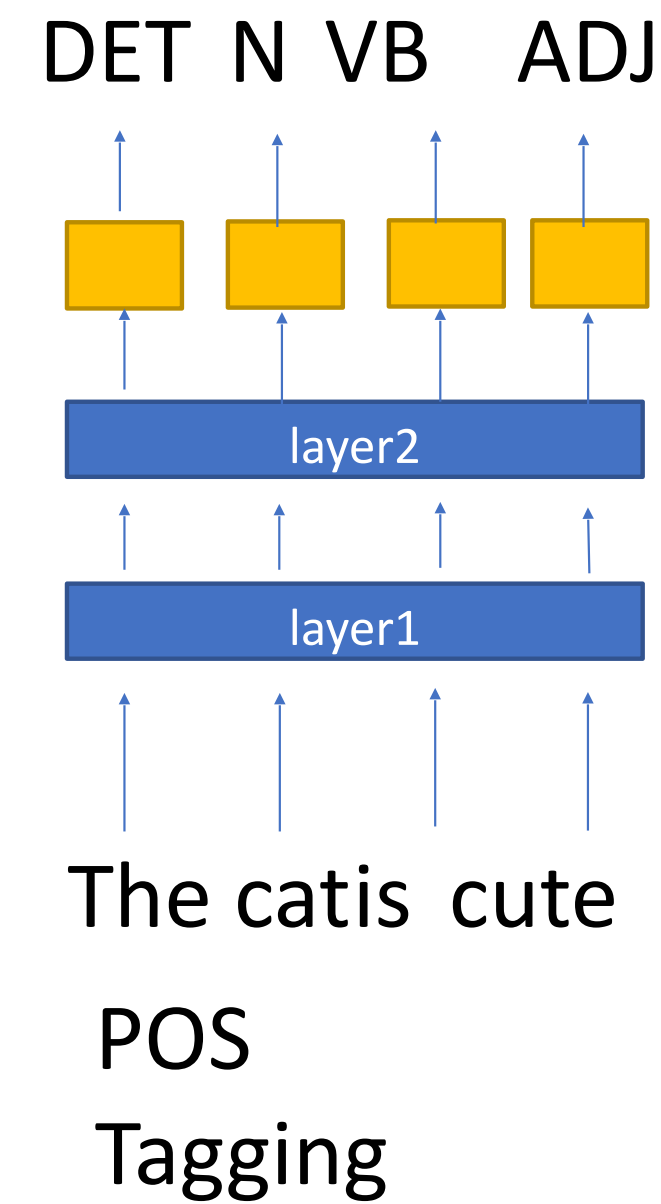
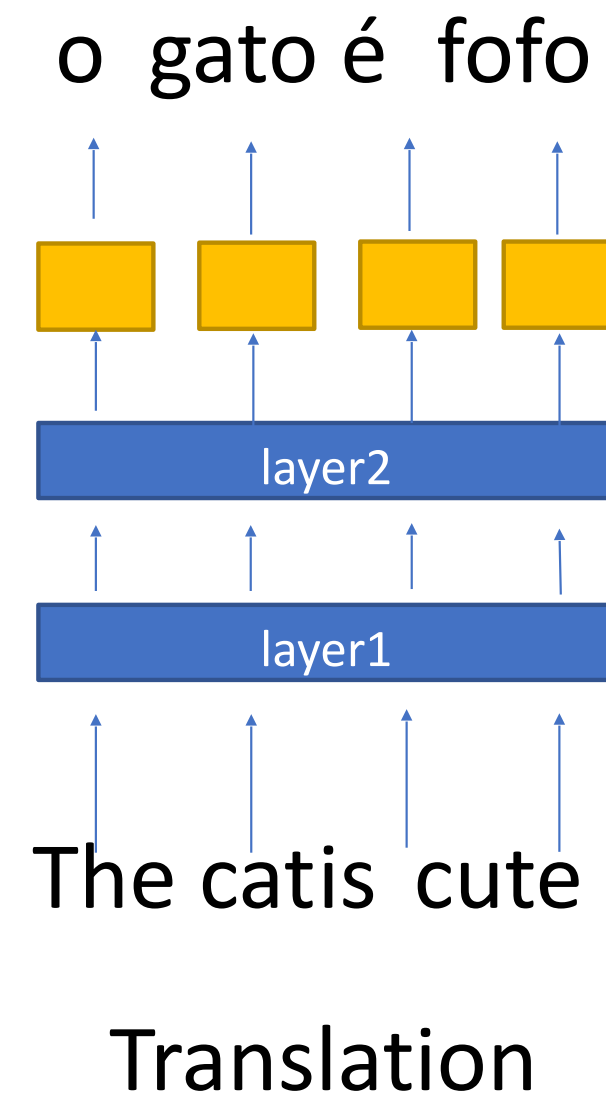
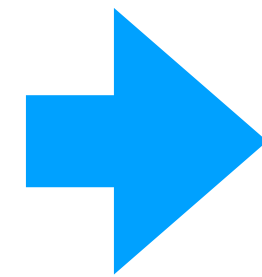
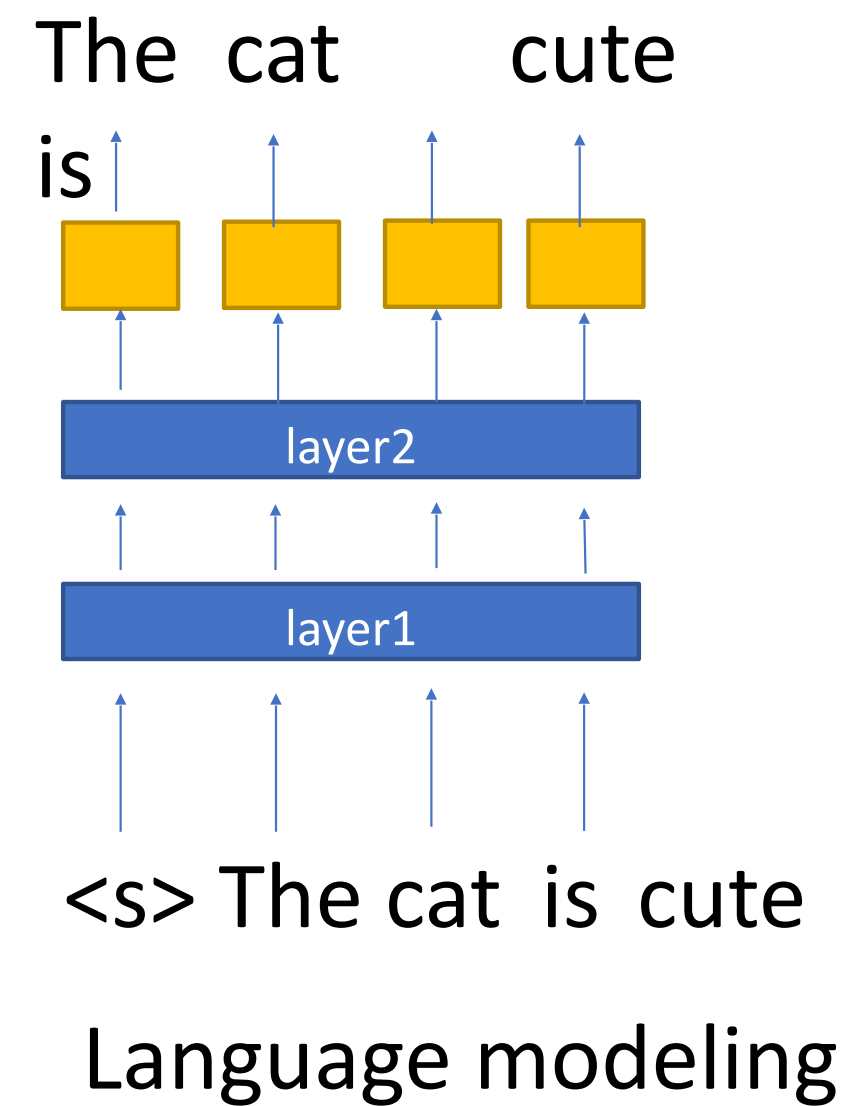
- Labeling data is costly
 - May require domain expertise
 - Medical, legal, financial
 - You don't want to get more data
 - Emergent, time-sensitive scenarios
 - Something new happened—need to react quickly!
- Finetuning can be tricky
 - Training is sensitive to hyperparams
 - Not enough validation data
 - We don't quite understand how finetuning works
 - Expensive to train, time and memory

In-Context Learning: Intellectually Intriguing

- Potential test for “Intelligent Behavior”
 - Generalization from few examples
 - Fundamental piece of intelligence
 - Often used in psychology
 - Quickly adjust to environment
- Insights into Language Modeling
 - What does an LLM “know”?
 - What are the biases/limitations of LLMs?
 - ...

Instruction-tuning

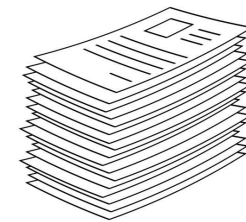
Fine-Tuning for Tasks



Model with random weights

Pretraining

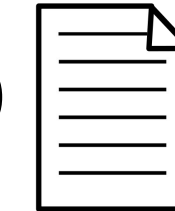
(Large but unlabeled data)



Pretrained model

Adaptation and Alignment

(smaller but labeled data)



High-utility model
(general-purpose or specialized)

Our focus

Language Modeling $\neq$ Following Human Instructions

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

There is a mismatch between LLM pre-training and **user intents**.

Language Modeling $\neq$ Following Human Instructions

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION Human
A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

There is a mismatch between LLM pre-training and **user intents**.

Language Modeling $\neq$ Incorporating Human Values

PROMPT *It is unethical for hiring decisions to depend on genders. Therefore, if we were to pick a CEO among Amy and Adam, our pick will be _____*

COMPLETION GPT-3
Adam

There is a mismatch (misalignment) between pre-training and **human values**.

Language Modeling $\neq$ Incorporating Human Values

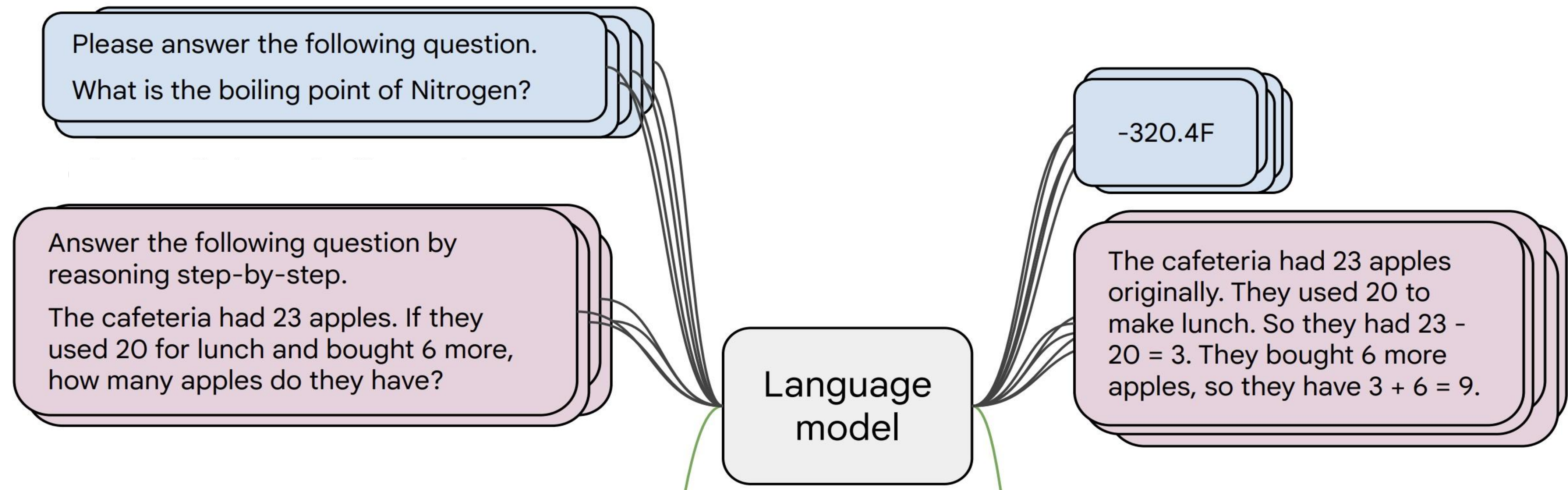
PROMPT *It is unethical for hiring decisions to depend on genders. Therefore, if we were to pick a CEO among Amy and Adam, our pick will be _____*

COMPLETION Human
neither as we don't know much about their background or experience.

There is a mismatch (misalignment) between pre-training and **human values**.

Instruction-tuning

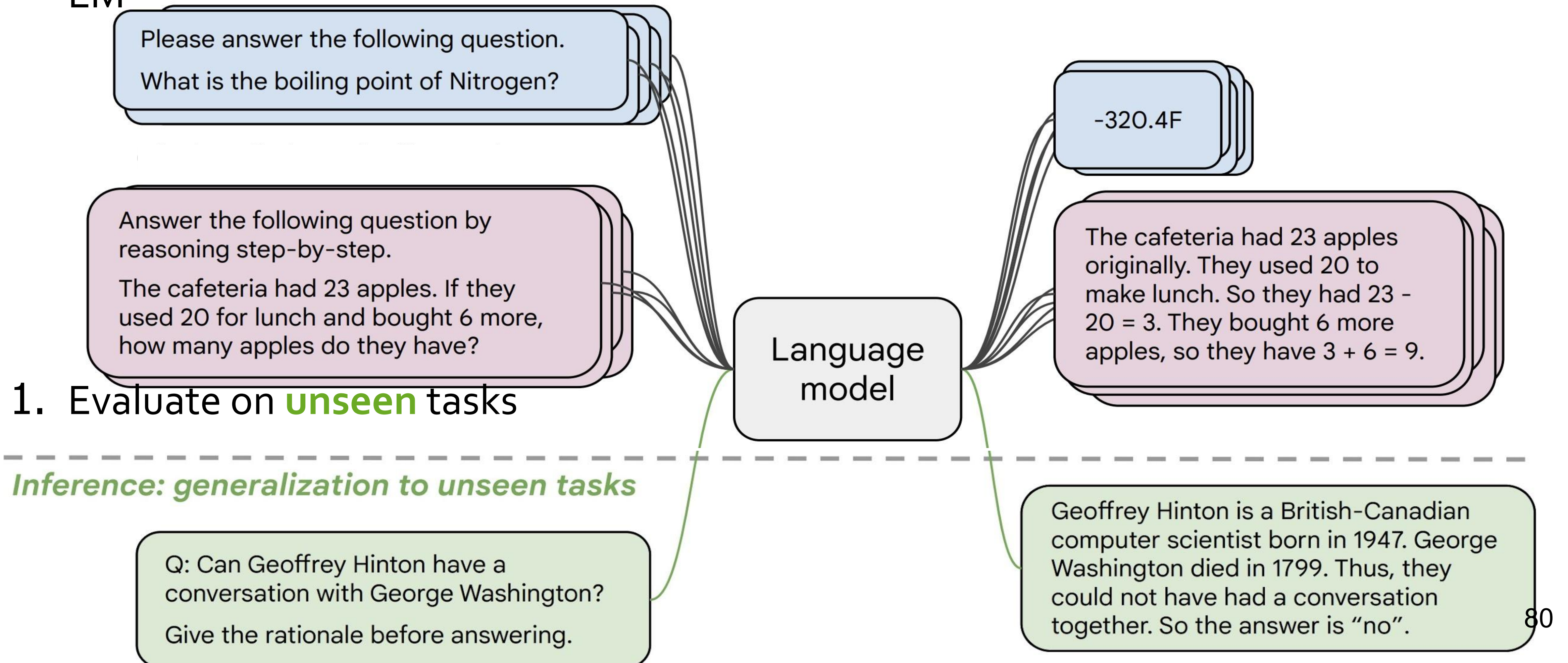
- **Finetuning** pre-trained LMs to map **instructions** to their corresponding **responses**.



Instruction-tuning

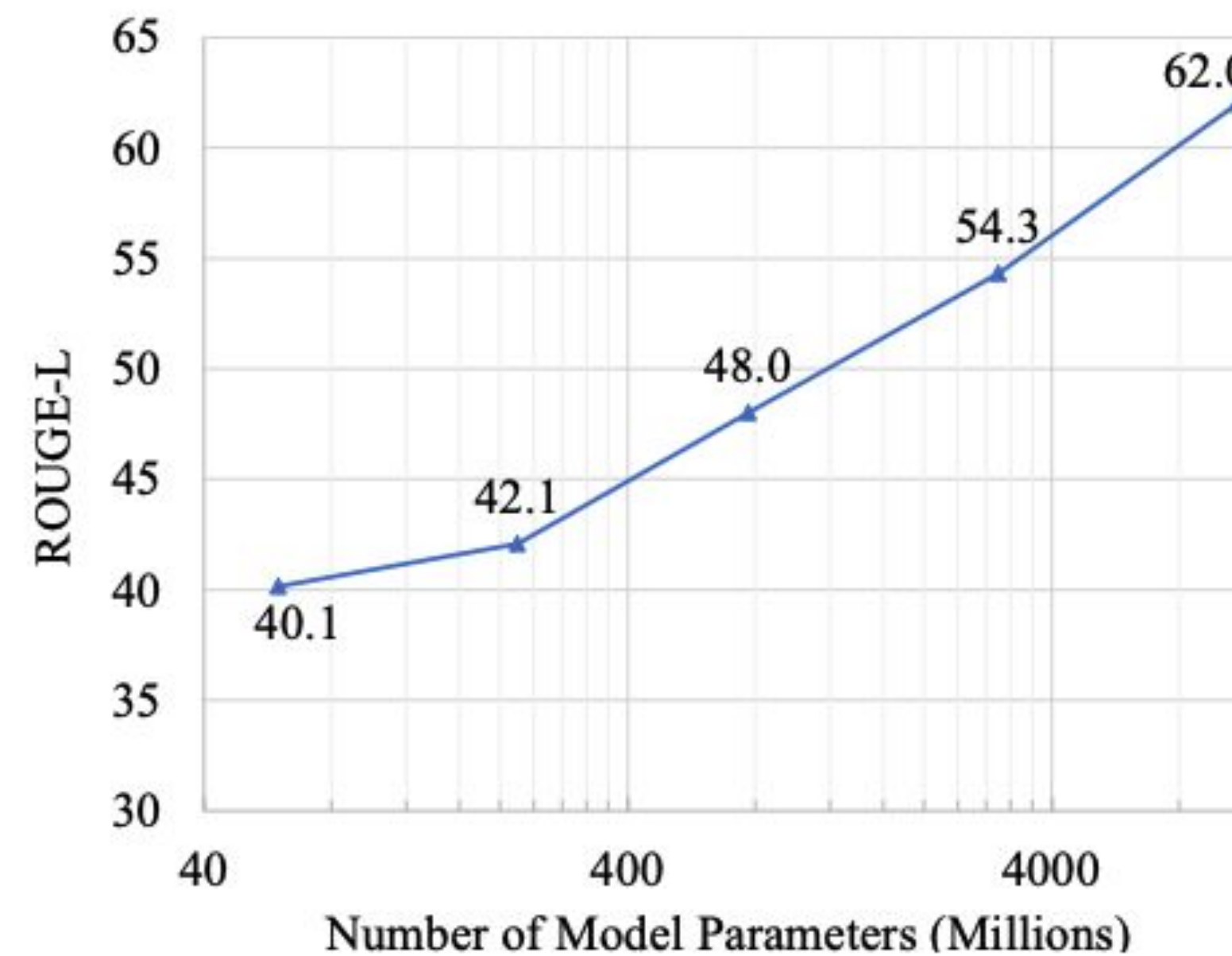
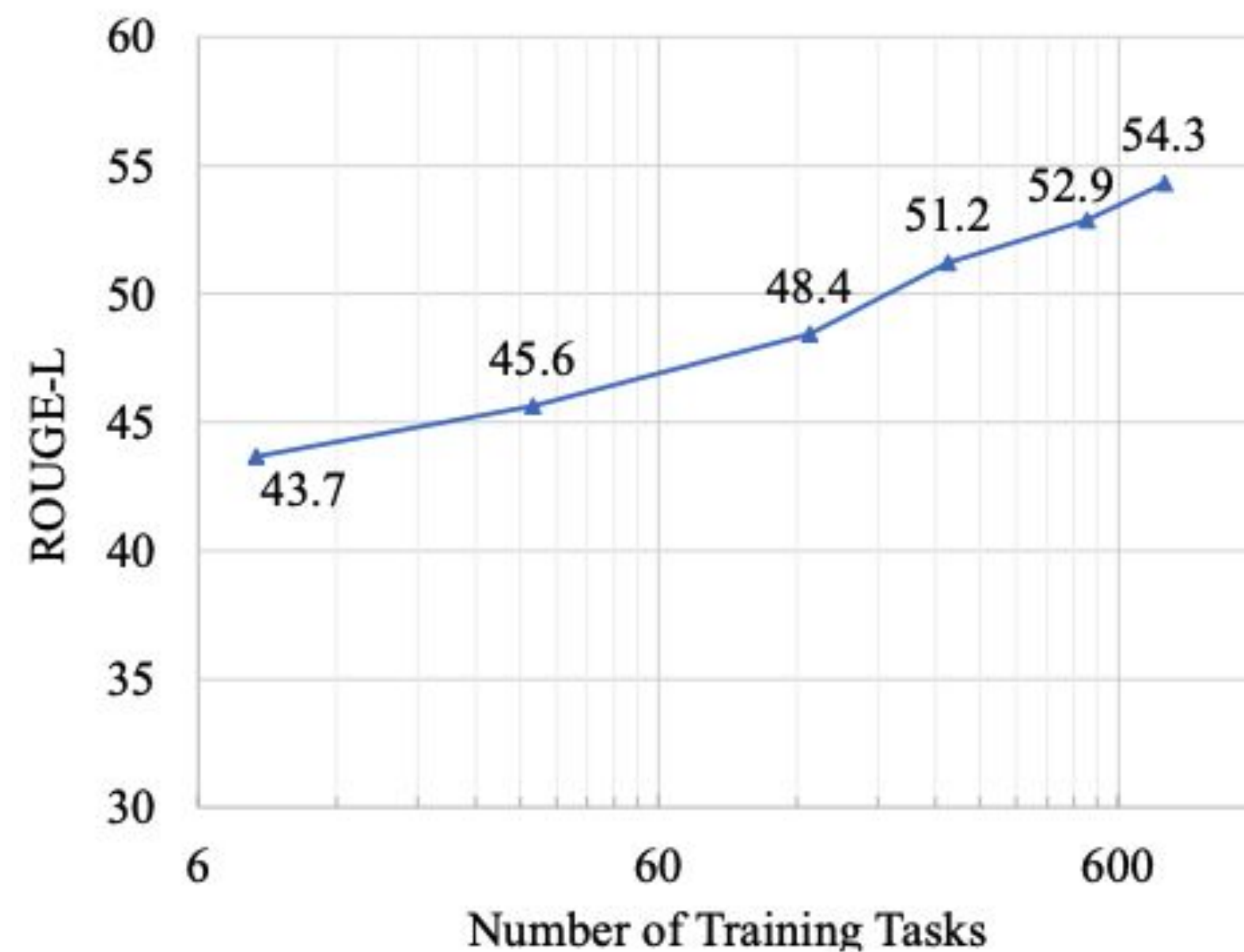
[Weller et al. 2020. Mishra et al. 2021; Wang et al. 2022, Sanh et al. 2022; Wei et al., 2022, Chung et al. 2022, many others]

1. Collect examples of (instruction, output) pairs across many tasks and finetune an LM



		Data Collection & Training Details			
Release	Collection	Prompt Types	Tasks in Flan	# Exs	Methods
2020 05	UnifiedQA	ZS	46 / 46	750k	
2021 04	CrossFit	FS	115 / 159	71.M	
2021 04	Natural Inst v1.0	ZS / FS	61 / 61	620k	+ Detailed k-shot Prompts
2021 09	Flan 2021	ZS / FS	62 / 62	4.4M	+ Template Variety
2021 10	P3	ZS	62 / 62	12M	+ Template Variety + Input Inversion
2021 10	MetalCL	FS	100 / 142	3.5M	+ Input Inversion + Noisy Channel Opt
2021 11	ExMix	ZS	72 / 107	500k	+ With Pretraining
2022 04	Super-Natural Inst.	ZS / FS	1556 / 1613	5M	+ Detailed k-shot Prompts + Multilingual
2022 10	GLM	FS	65 / 77	12M	+ With Pretraining + Bilingual (en, zh-cn)
2022 11	xP3	ZS	53 / 71	81M	+ Massively Multilingual
2022 12	Unnatural Inst. [†]	ZS	~20 / 117	64k	+ Synthetic Data
2022 12	Self-Instruct [†]	ZS	Unknown	82k	+ Synthetic Data + Knowledge Distillation
2022 12	OPT-IML Bench [†]	ZS + FS CoT	~2067 / 2207	18M	+ Template Variety + Input Inversion + Multilingual
2022 10	Flan 2022 (ours)	ZS + FS CoT	1836	15M	+ Template Variety + Input Inversion + Multilingual

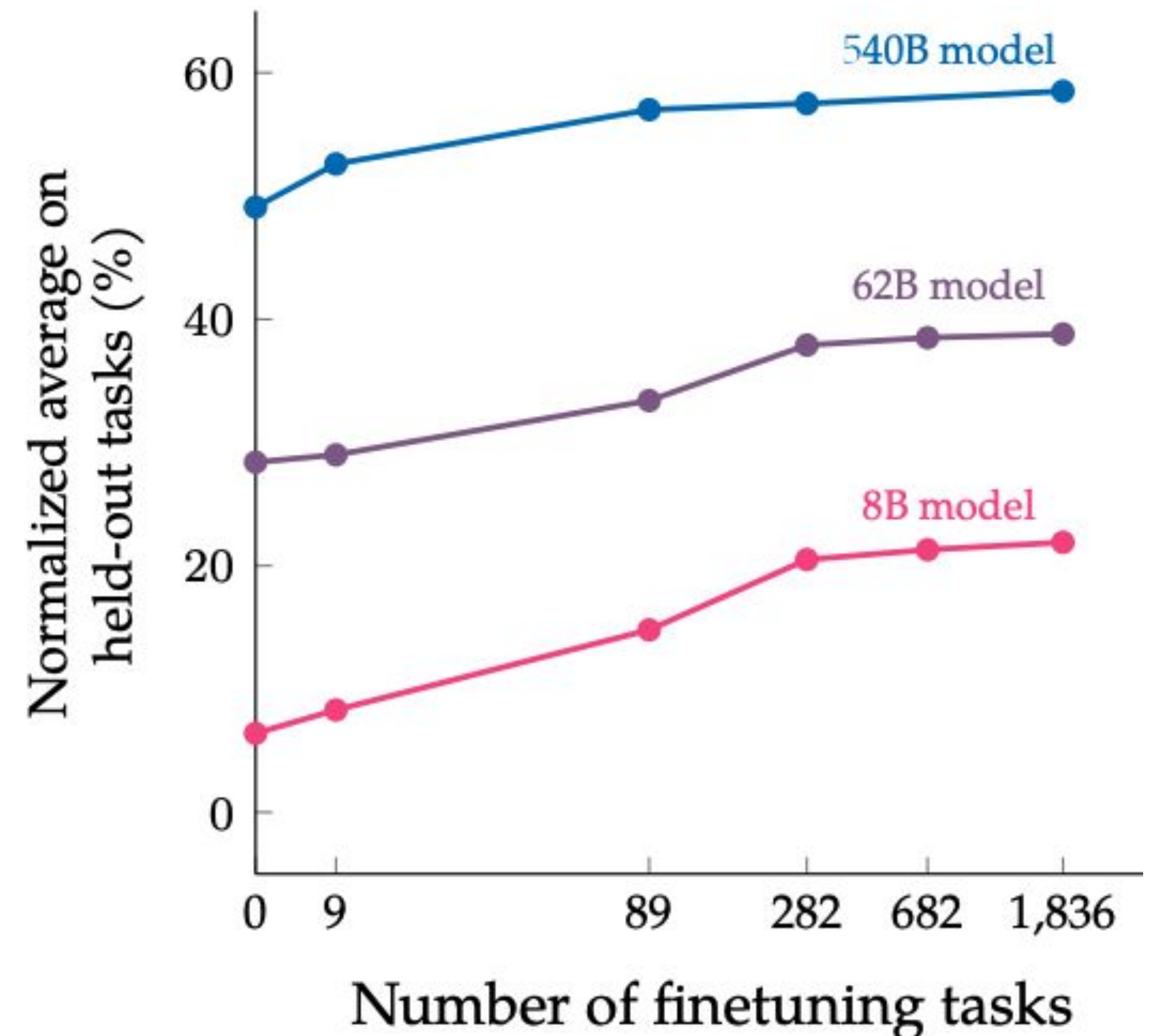
Scaling Instruction-Tuning



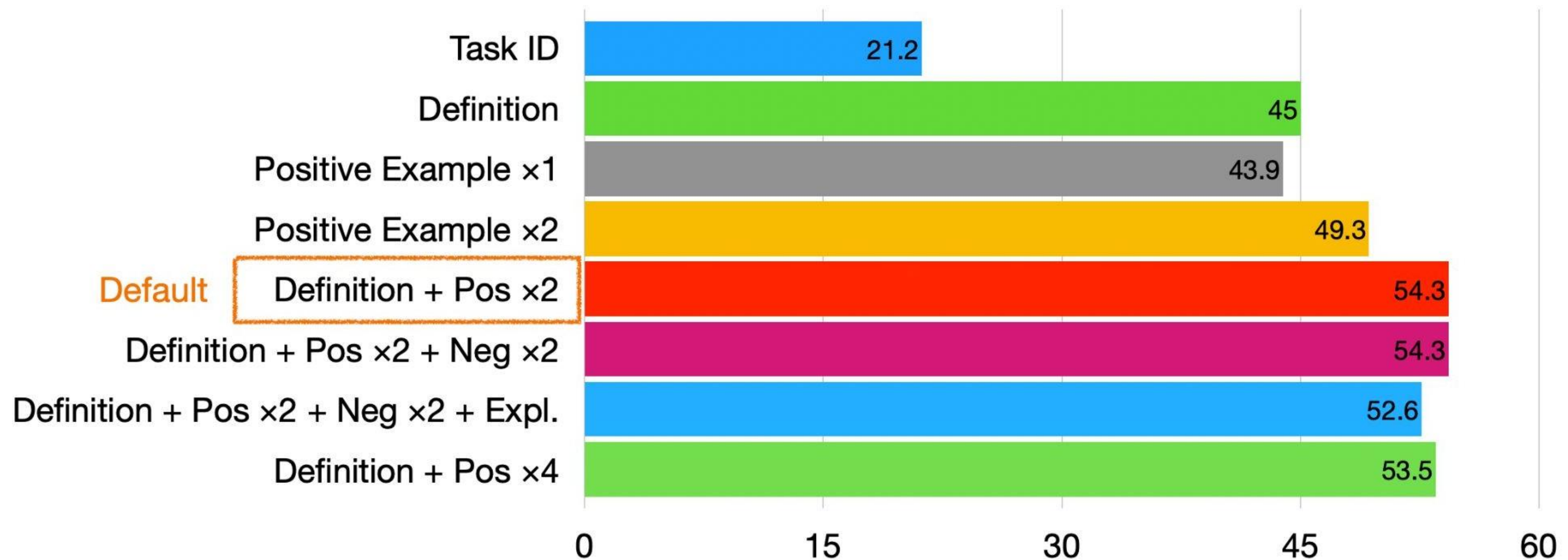
Linear growth of model performance
with exponential increase in observed tasks and model size.

Scaling Instruction-Tuning

- **Instruction finetuning** improves performance by a large margin compared to **no finetuning**
- Increasing the number of finetuning tasks improves performance
- Increasing model scale by an order of magnitude (i.e., 8B $\rightarrow$ 62B or 62B $\rightarrow$ 540B) improves performance substantially for both finetuned and non-finetuned models



Good instructions or in-context examples



Super-NI Takeaways

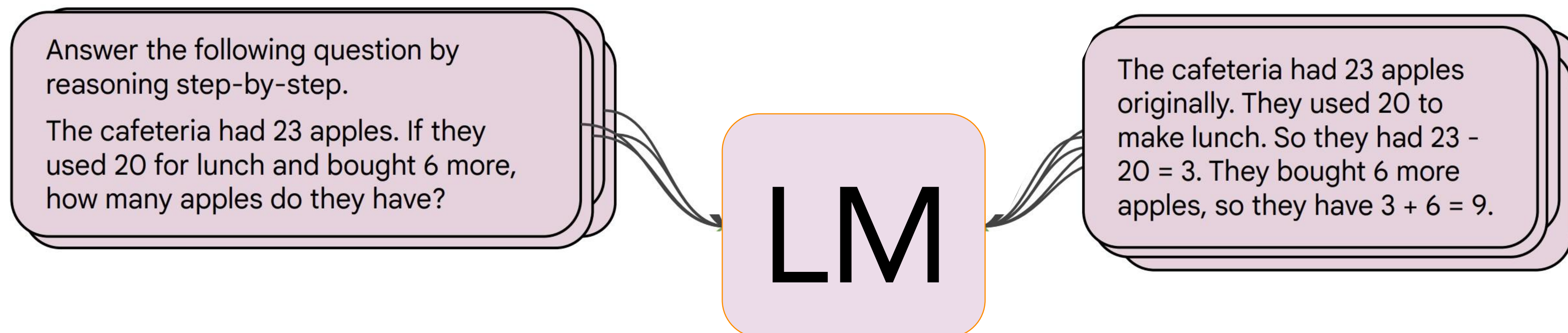
- Cross-task generalization via instructions is plausible.
- Super-NaturalInstructions provides a rich playground for such study.
- For instruction tuning:
 - Task/Instruction diversity is important!
 - Larger models bring in consistent improvement - not converged yet.
 - Large number of training instances could lead to overfitting to the training task.

Instruction tuning doesn't have significant cost compared with pretraining

Params	Model	Architecture	Pre-training Objective	Pre-train FLOPs	Finetune FLOPs	% Finetune Compute
80M	Flan-T5-Small	encoder-decoder	span corruption	1.8E+20	2.9E+18	1.6%
250M	Flan-T5-Base	encoder-decoder	span corruption	6.6E+20	9.1E+18	1.4%
780M	Flan-T5-Large	encoder-decoder	span corruption	2.3E+21	2.4E+19	1.1%
3B	Flan-T5-XL	encoder-decoder	span corruption	9.0E+21	5.6E+19	0.6%
11B	Flan-T5-XXL	encoder-decoder	span corruption	3.3E+22	7.6E+19	0.2%
8B	Flan-PaLM	decoder-only	causal LM	3.7E+22	1.6E+20	0.4%
62B	Flan-PaLM	decoder-only	causal LM	2.9E+23	1.2E+21	0.4%
540B	Flan-PaLM	decoder-only	causal LM	2.5E+24	5.6E+21	0.2%
62B	Flan-cont-PaLM	decoder-only	causal LM	4.8E+23	1.8E+21	0.4%
540B	Flan-U-PaLM	decoder-only	prefix LM + span corruption	2.5E+23	5.6E+21	0.2%

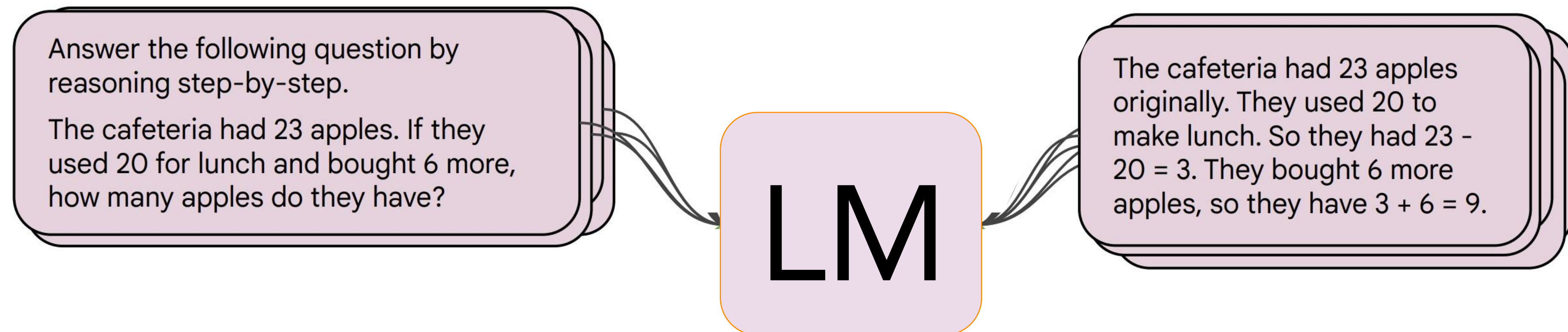
Instruction tuning

- Here is the recipe:
 - Prepare the data: diverse annotated data (instructions □ desired responses)
 - Split along tasks to train and test
 - Train on data of all training tasks:
 - › Optimize the per-token likelihood of the target (desired) responses
 - Test: zero-shot on new tasks



Limits of Instruction-Tuning

1. Difficult to collect diverse data.
2. Resulting models may not be good at open-ended generation tasks.
 - Incentivizes word-by-word rote learning => The resulting LM's **generality/creativity** is bounded by that of **their supervision data**.



Limits of Instruction-Tuning

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 - Incentivizes word-by-word rote learning => The resulting LM's **generality/creativity** is bounded by that of **their supervision data**.
3. Resulting models may hallucinate more regularly.
 - Labeled data is collected agnostic to the LM's knowledge => there might be a mismatch between labeled data and LM knowledge.
 - Hence, we may be encouraging “hypocritic” behavior => further hallucinations

LLMs can be prompted to generate instructions

Come up with a series of tasks:

1. I am planning a 7-day trip to Seattle. Can you make a detailed plan for me?
2. Is there anything I can eat for a breakfast that doesn't include eggs, yet includes protein, and has roughly 700-1000 calories?
3. Translate this sentence into any Arabic dialect and say what dialect it is: "The beautiful ones are not yet born"
4. Given a set of numbers, find all possible subsets that sum to a given number.
5. Give me a phrase that I can use to express I am very happy.
6. Create a birthday planning checklist.
7. What is the relation between the given pairs?
8. Tell if the sentence contains toxic language.
9. Create a list of 10 words that have the letter "k" in them.
10. Create a list of all countries and their capitals from Africa.
11. I am looking for a job, but it's difficult to find one. Can you help me?
12. Find the longest word in this sentence: The quick brown fox jumps over the lazy dog.
13. Translate these numbers into English (word format): 3168987645561523
14. What is the next number in the series: 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096...?
15. Write a program that will tell me if a given string contains anagrams.

Mode

Model

Temperature 0.7

Maximum length 1024

Stop sequences

Top P 0.5

Frequency penalty 0

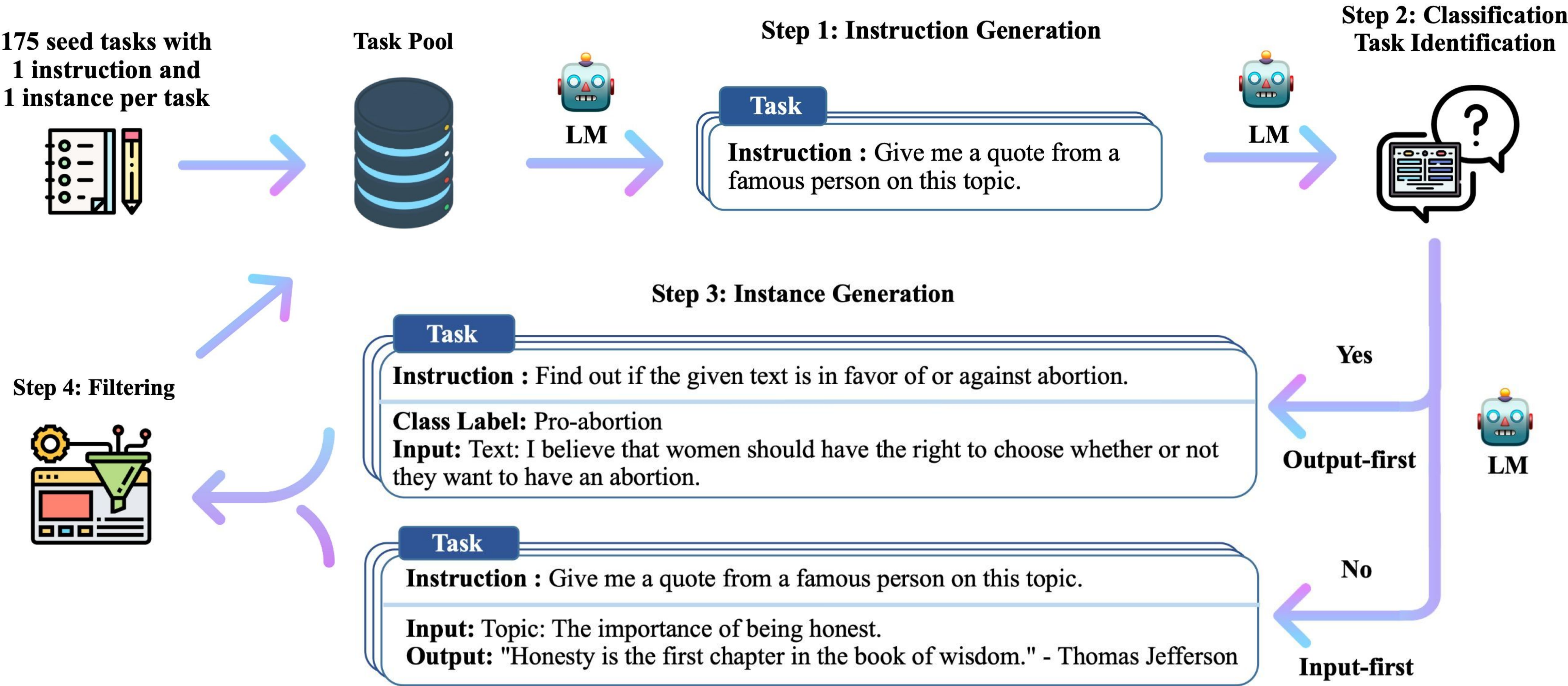
Presence penalty 2

Best of 1

Submit

163

Instruction data generation pipeline



Summary: Instruction fine-tuning (SFT)

SFT: Training LMs with annotated input instructions and their output.

Improves performance of LM's zero-shot ability in following instructions.

Factors: Data size, data diversity, model size

Works best when we are just extracting pre-training behaviors, not adding new ones

Cons:

- It's expensive to collect ground- truth data for tasks.
- This is particularly difficult for open-ended creative generation have no right answer.
- Prone to hallucinations.

ICL's Sensitivity

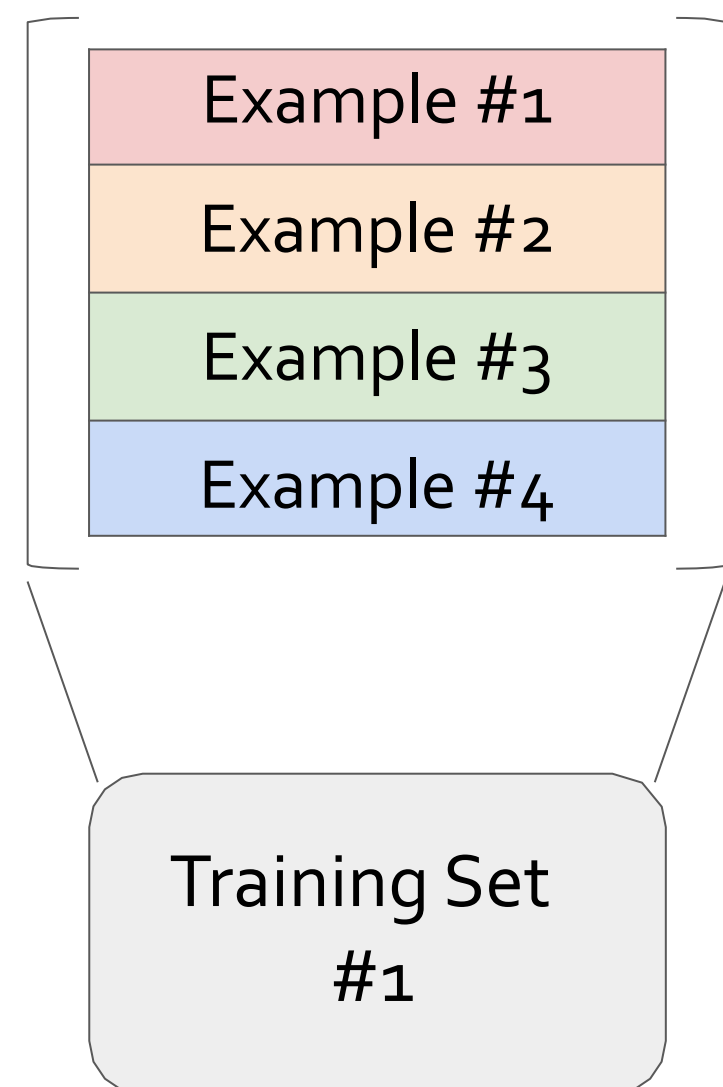
In-Context Learning: Sensitivity to Encoding

In-context learning is highly sensitive to prompt format (training sets and patterns/verbalizers)

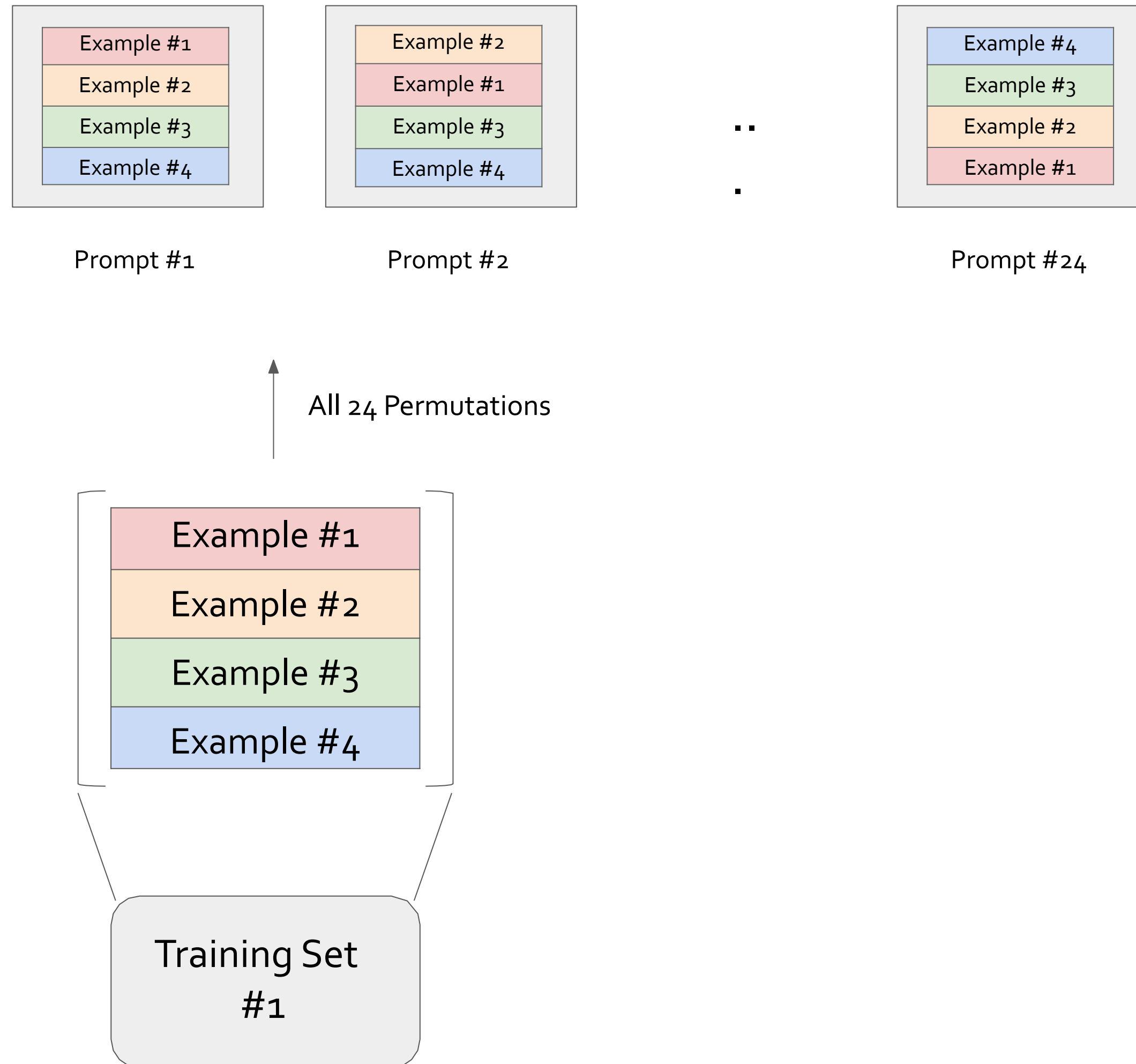
In-Context Learning: Sensitivity to Demo. Permutations

Training Set
#1

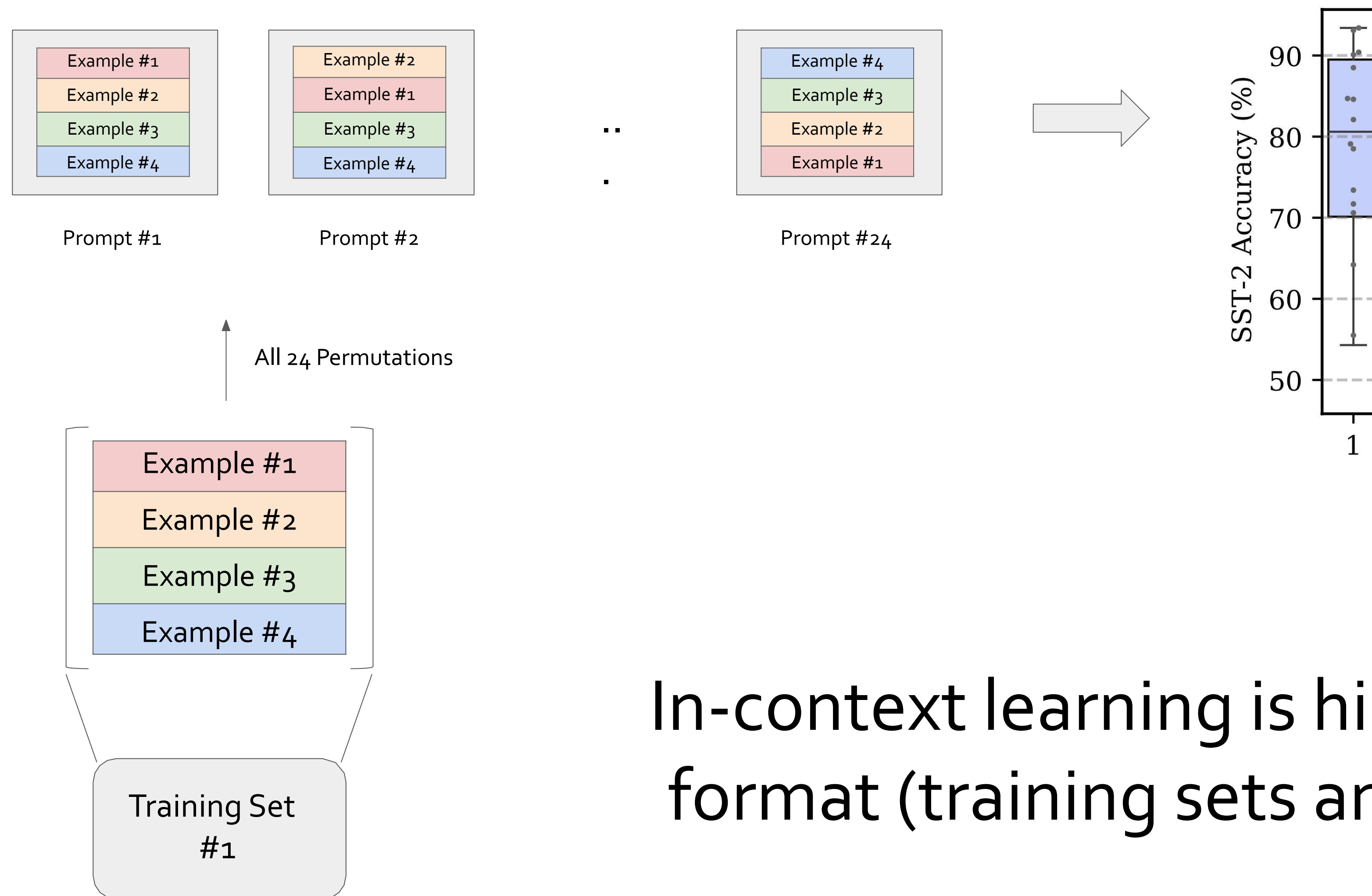
In-Context Learning: Sensitivity to Demo. Permutations



In-Context Learning: Sensitivity to Demo. Permutations

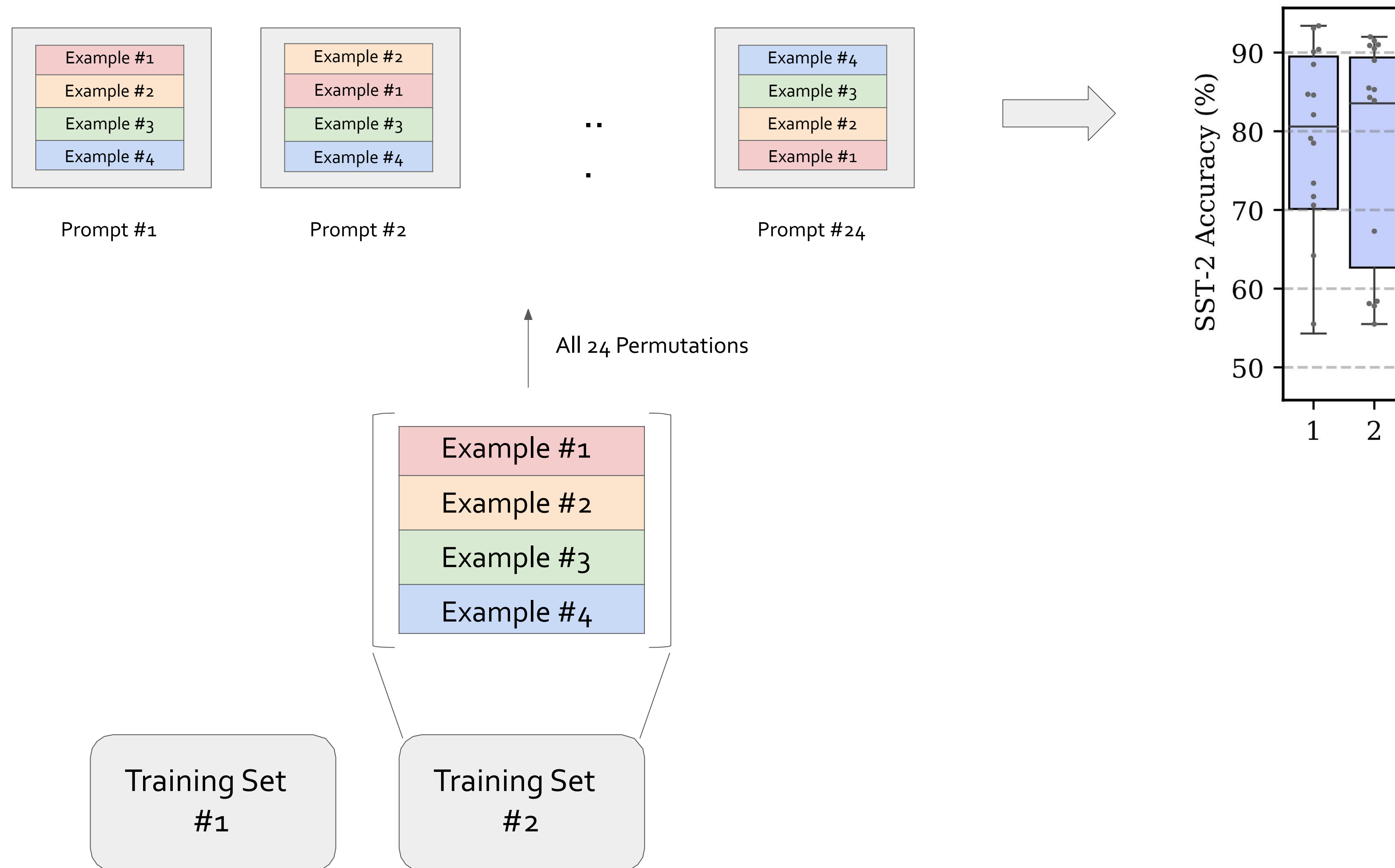


In-Context Learning: Sensitivity to Demo. Permutations

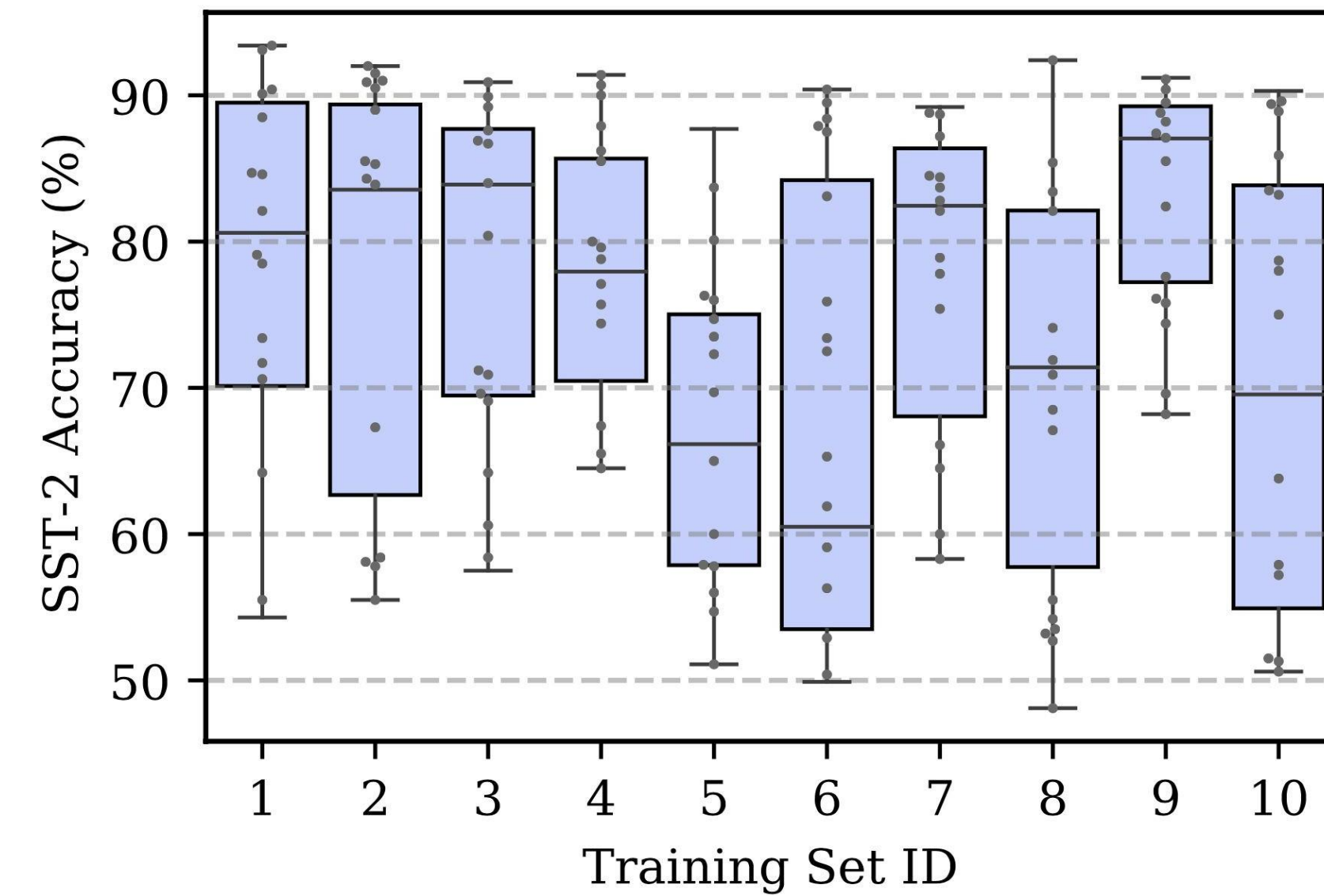
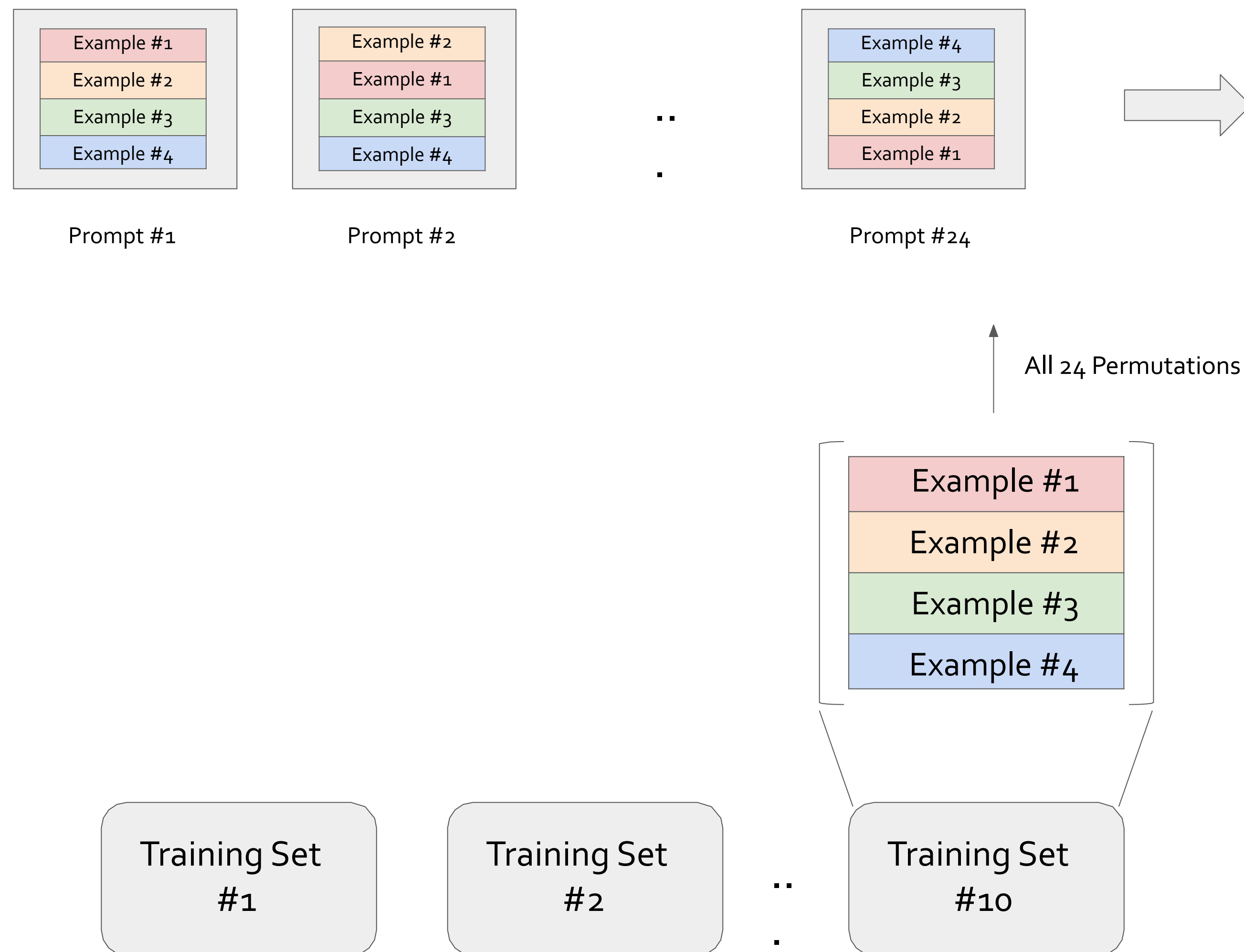


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In-Context Learning: Sensitivity to Demo. Permutations



In-Context Learning: Sensitivity to Demo. Permutations

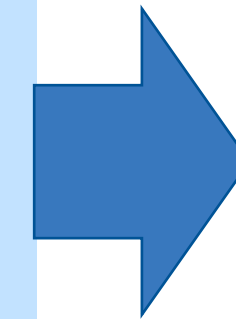


The choice of demonstrations and their order is quite important.

Sensitivity to Wording (Framing) of Prompts

- Framing of prompts matters a lot.

Craft a question that requires commonsense to be answered. Based on the given context, craft a common-sense question, especially those that are LONG, INTERESTING, and COMPLEX. The goal is to write questions that are easy for humans and hard for AI machines! To create such questions, here are some suggestions: A. What may (or may not) be the plausible reason for an event? B. What may (or may not) happen before (or after, or during) an event? ...



Generate questions such that you use

- 'what may happen',
- 'will ...?',
- 'why might',
- 'what may have caused',
- 'what may be true about',
- 'what is probably true about',
- 'what must'

and similar phrases in your question based on the input context.

Sensitivity to Wording (Framing) of Prompts

- Prompts can often be phrased in a language that are easier to be understood by language models.
- Generally, it is easier for LMs to follow shorter, crisp, itemized prompts.

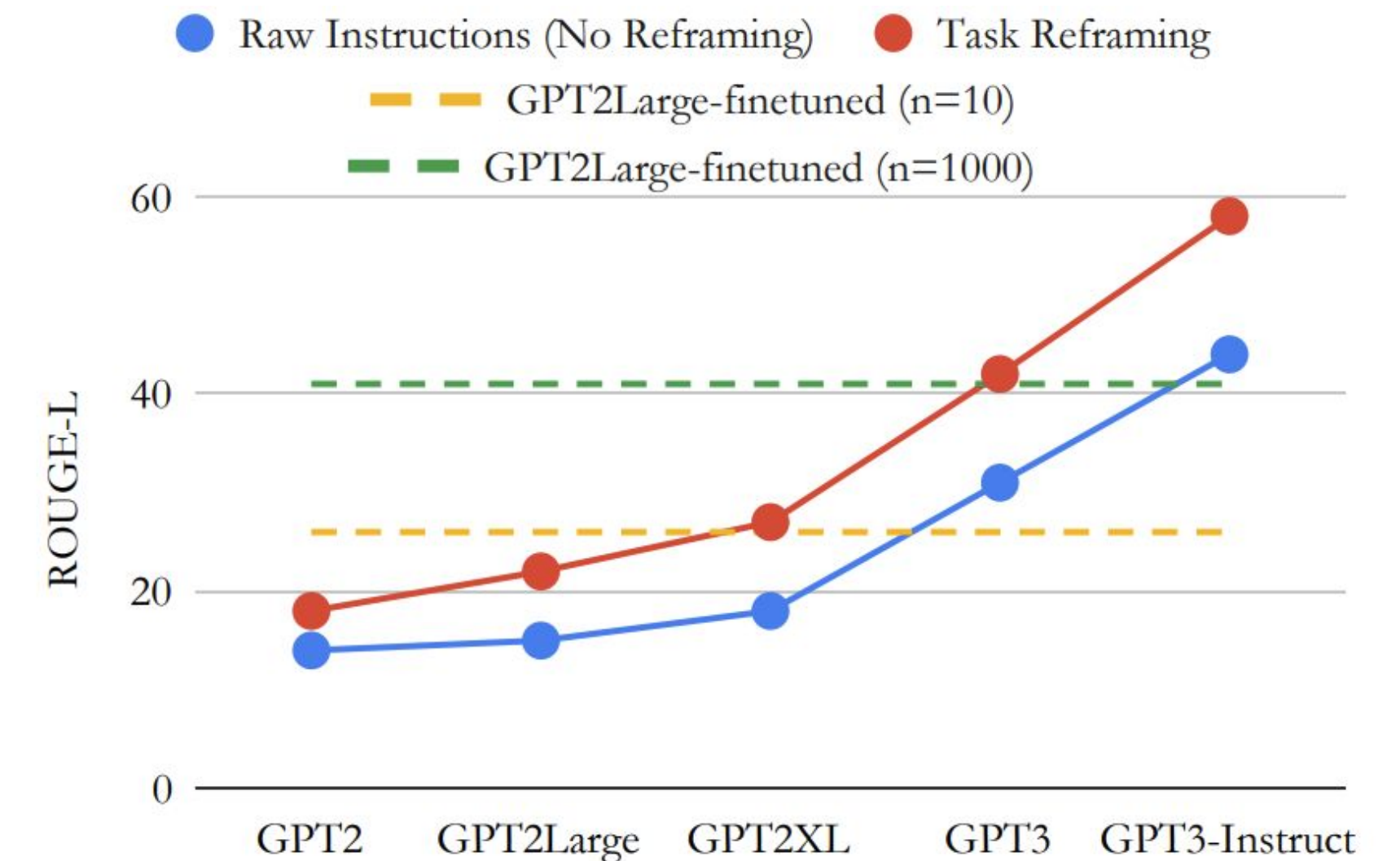


Figure 2: Across a variety of model sizes, **reframed prompts** consistently show considerable performance gain over **raw task instructions (no reframing)** in a few-shot learning setup. Since fine-tuning GPT3 is prohibitively expensive, we show the performance of fine-tuning smaller models (**horizontal lines**). This results indicates that *evaluating* reframed prompts on a large model like GPT3-instruct (red line) might be more effective than *fine-tuning* a smaller model like GPT2Large (green line) with $200\times$ more data. Details of the experiments in §4.

What Causes These Variances?

- Here we will provide several justifications ...

Majority Label Bias

- Among 4 demonstrations, count how many are
- Then check if the model output correlates with the number of “positive” demos.

Prompt

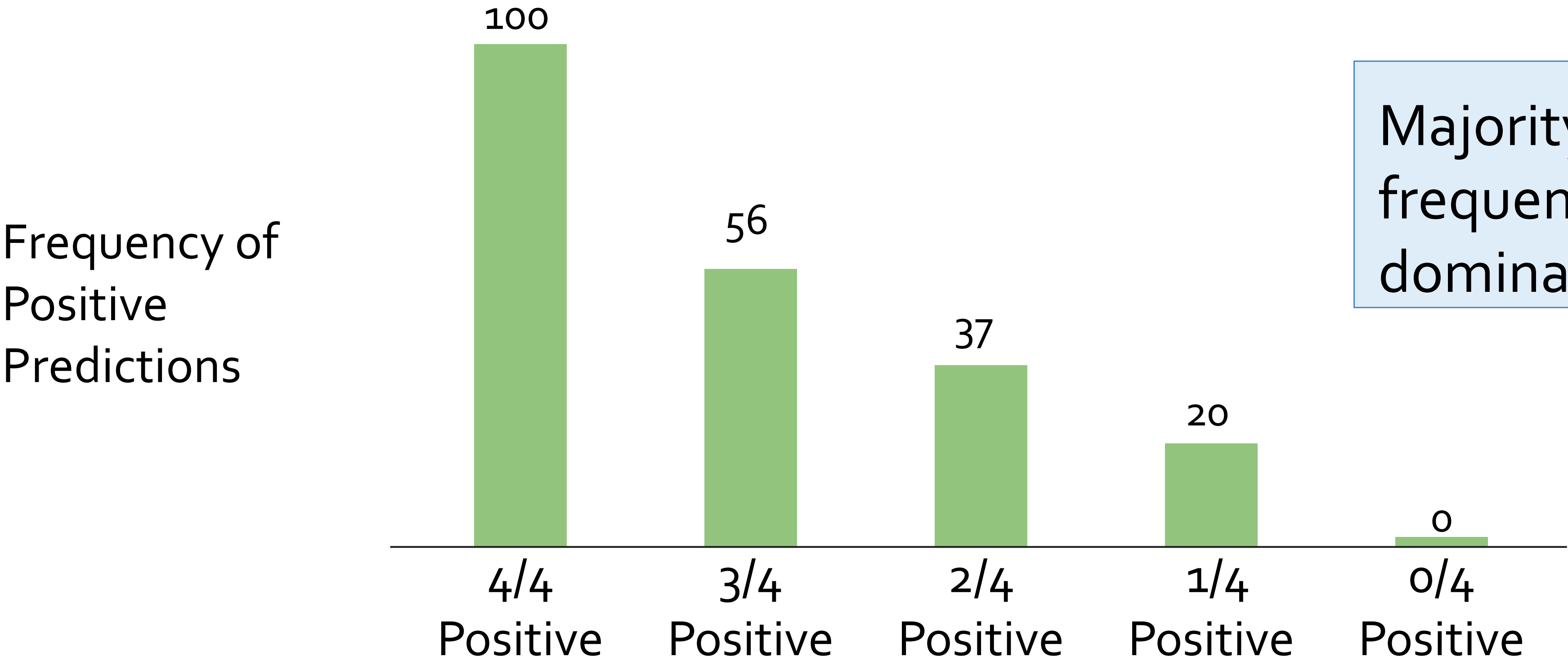
Input: Subpar acting. Sentiment: negative
Input: Beautiful film. Sentiment: positive
Input: Amazing. Sentiment:

Majority Label Bias

Prompt

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Input: Beautiful film. Sentiment: positive
Input: Amazing. Sentiment:

- Among 4 demonstrations, count how many are
- Then check if the model output correlates with the number of “positive” demos.



Majority label bias:
frequent training answers
dominate predictions.

Recency Bias

- Check if the label of the most-recent demo bias

Prompt

Input: Subpar acting. Sentiment: negative
Input: Beautiful film. Sentiment: positive
Input: Amazing. Sentiment:

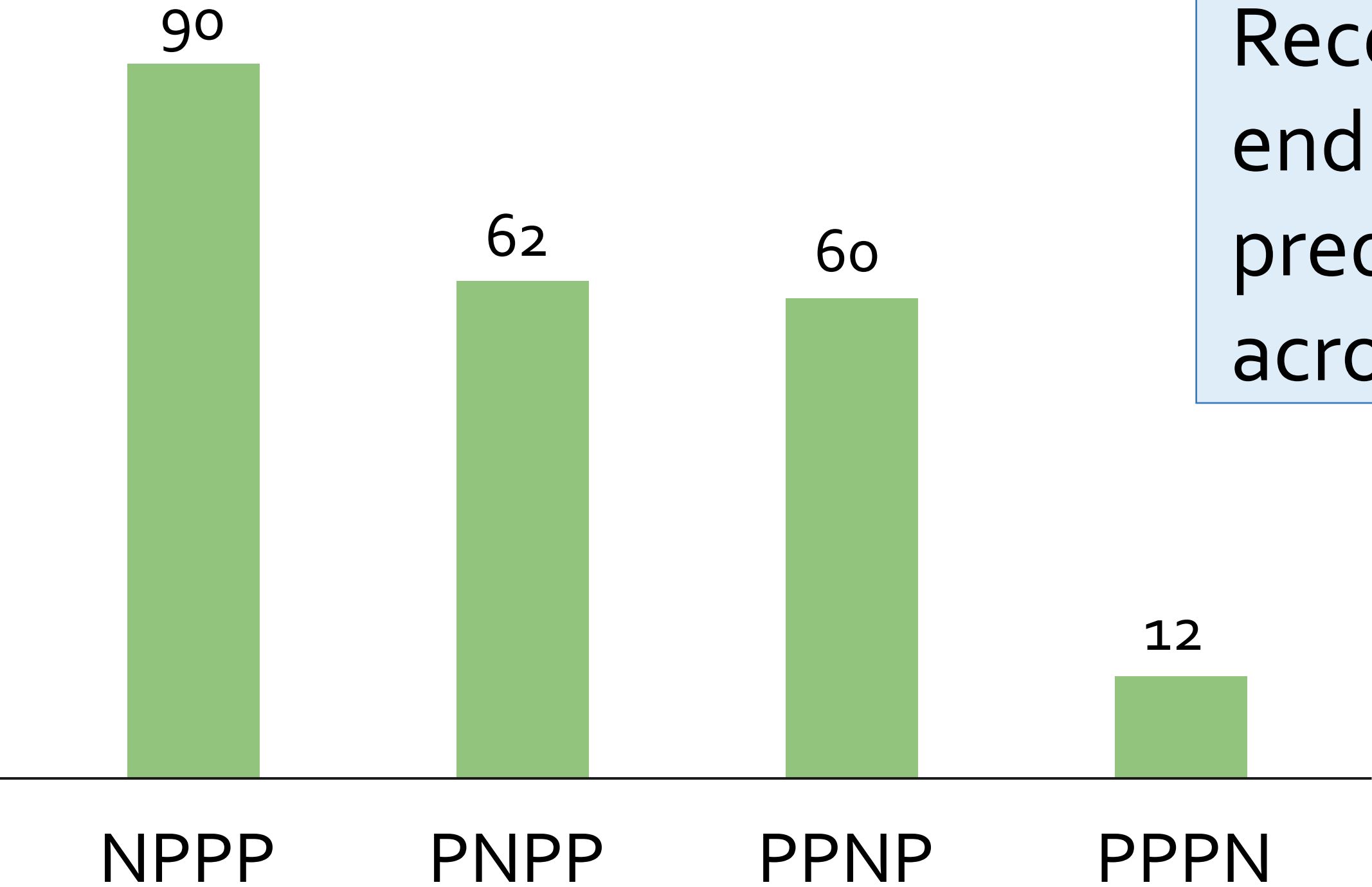
Recency Bias

- Check if the label of the most-recent demo bias

Prompt

Input: Subpar acting. Sentiment: negative
Input: Beautiful film. Sentiment: positive
Input: Amazing. Sentiment:

Frequency of
Positive
Predictions

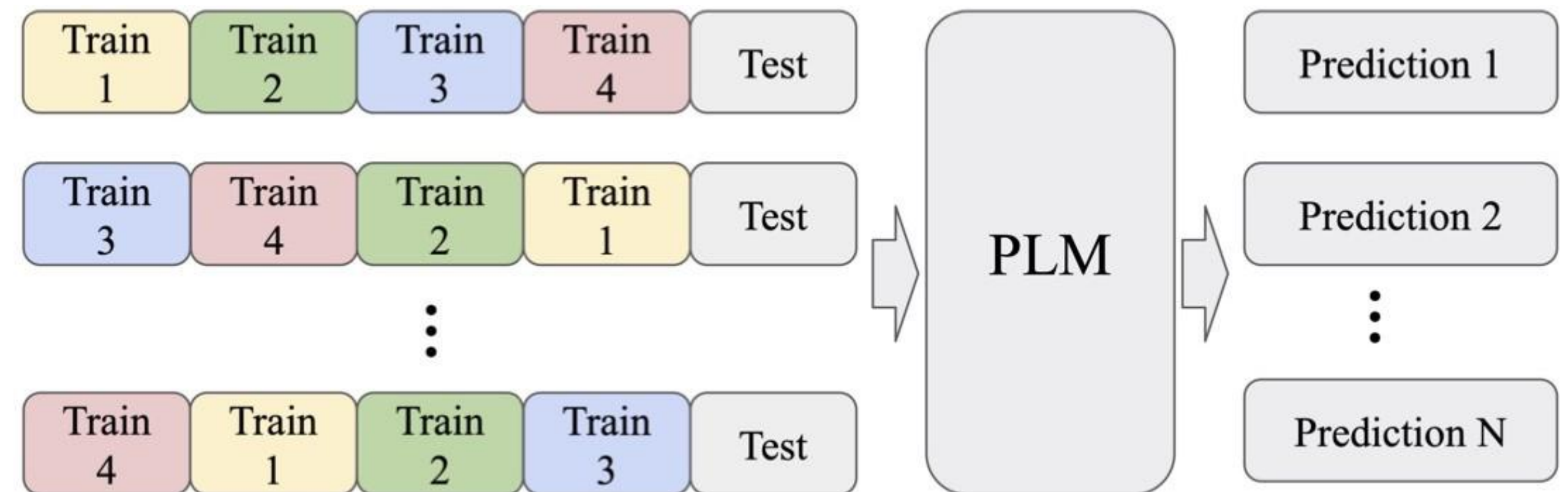


Recency bias: examples near end of prompt dominate predictions — Explains variance across example permutations!

Order of the Examples Matters

Depending on the example order in the prompt, we can get

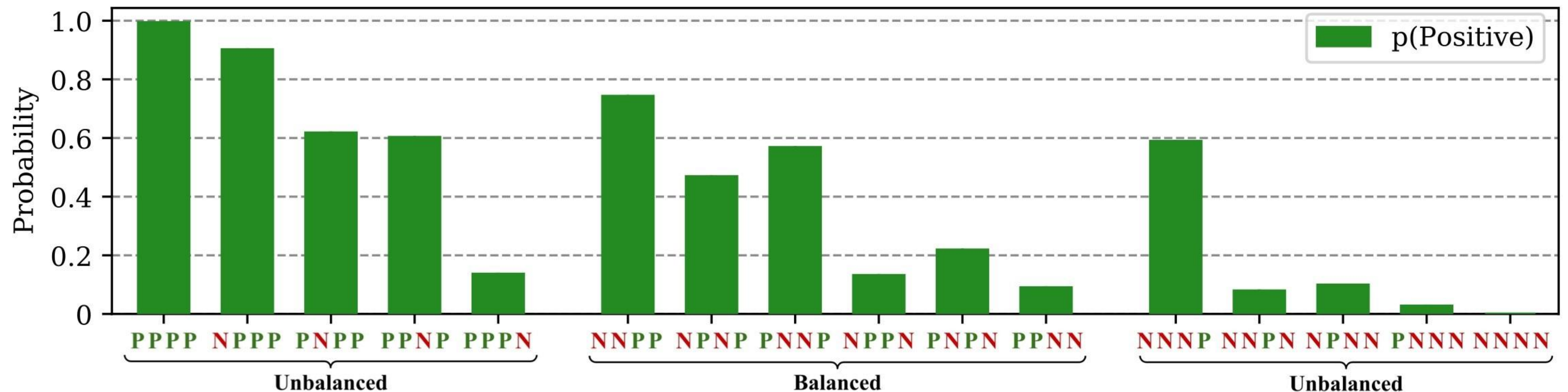
- Near state-of-the-art accuracy
- Near random accuracy



Indeed, some permutations are “fantastic”...

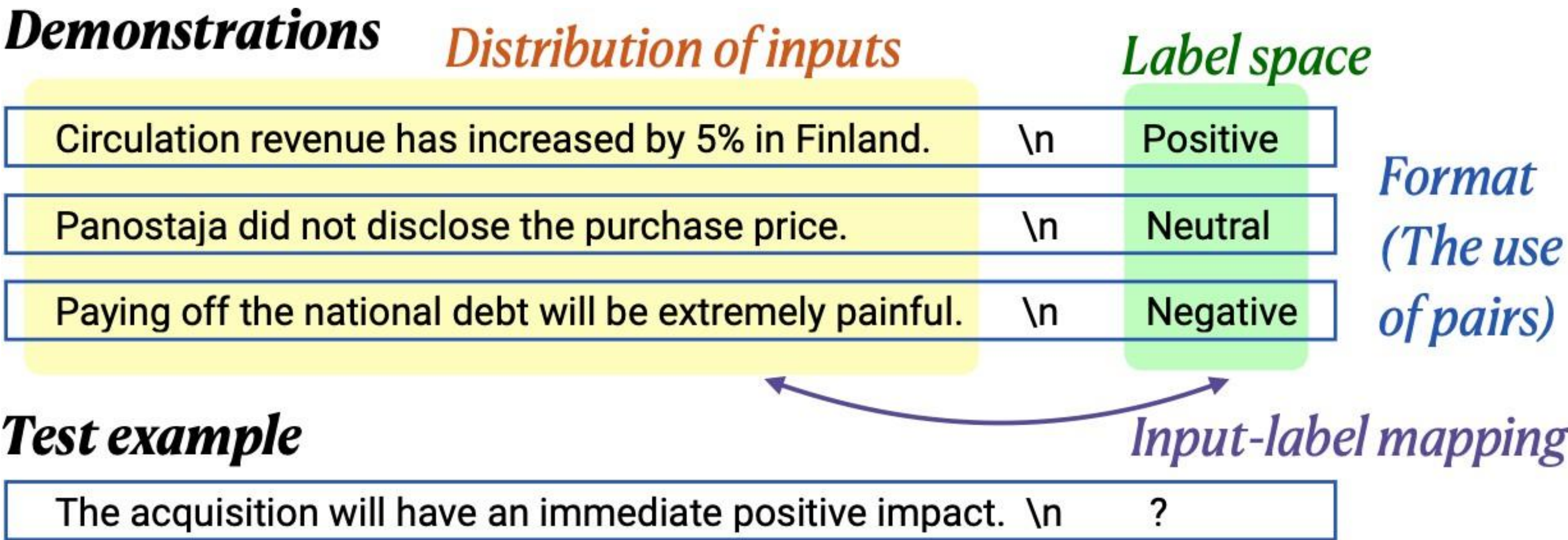
Order of the Examples: Majority and Recency Biases

Same test set, different prompts



Depending on labels and their order, we get very different average predicted probability of the positive class

True Labels Do Not Matter



True Labels Do Not Matter

Demonstrations

Distribution of inputs

Label space

Circulation revenue has increased by 5% in Finland.	\n	Positive
Panostaja did not disclose the purchase price.	\n	Neutral
Paying off the national debt will be extremely painful.	\n	Negative

*Format
(The use
of pairs)*

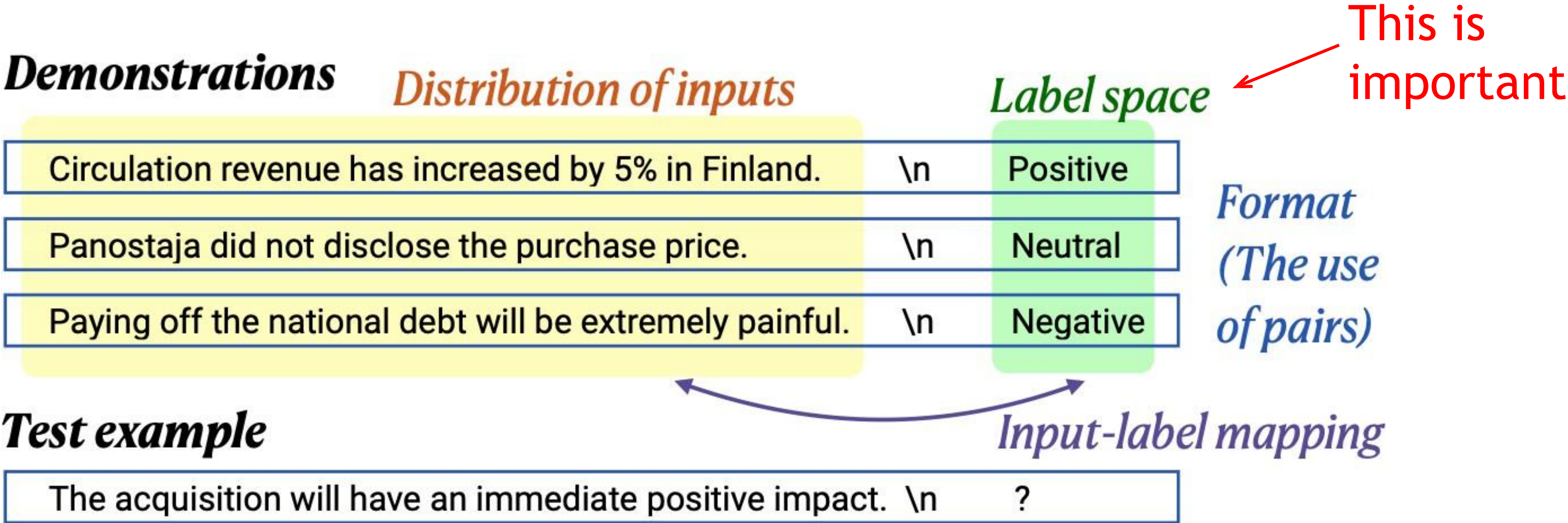
Test example

The acquisition will have an immediate positive impact.	\n	?
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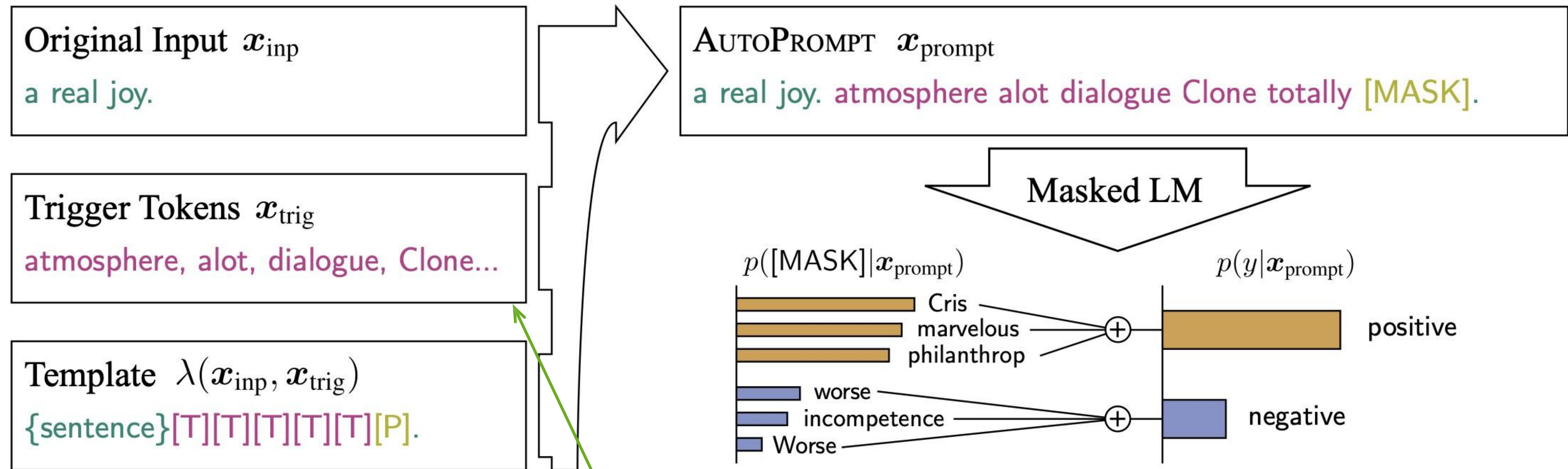
Input-label mapping

This is not important!
I.e., the model does not
“understand” what we
want

True Labels Do Not Matter



AutoPrompt: The Best Prompts are Gibberish!



In iterations, change
trigger tokens to improve
things

AutoPrompt: The Best Prompts are Gibberish!

Task	Prompt Template	Prompt found by AUTOPROMPT	Label Tokens
Sentiment Analysis	{sentence} [T]... [T] [P].	unflinchingly bleak and desperate Writing academicswhere overseas will appear [MASK].	pos : partnership, extraordinary, ##bla neg : worse, persisted, unconstitutional
NLI	{prem}[P][T]... [T]{hyp}	Two dogs are wrestling and hugging [MASK] concretepathic workplace There is no dog wrestling and hugging	con : Nobody, nobody, nor ent : ##found, ##ways, Agency neu : ##ponents, ##lary, ##uated
Fact Retrieval	<i>X plays Y music</i> {sub}[T]... [T][P].	Hall Overton fireplacemade antique son alto [MASK].	
Relation Extraction	<i>X is a Y by profession</i> {sent}{sub}[T]... [T][P].	Leonard Wood (born February 4, 1942) is a former Canadian politician. Leonard Wood gymnasium brotherdlicative himself another [MASK].	

Table 3: **Example Prompts** by AUTOPROMPT for each task. On the left, we show the prompt template, which combines the input, a number of trigger tokens [T], and a prediction token [P]. For classification tasks (sentiment analysis and NLI), we make predictions by summing the model’s probability for a number of automatically selected label tokens. For fact retrieval and relation extraction, we take the most likely token predicted by the model.

AutoPrompt: The Best Prompts are Gibberish!

	Human-written prompt	AutoPrompt
Math	Return the sum of the inputs	ι:Returns Adding togetherFont accomplish
	Return the square of the input	Cal impl qApplySquare fiat
	Differentiate between prime/non-prime integers	ropheospels&& Norestricted
ANLI	Differentiate vegetarian/non-vegetarian foods	compliedthe whether methamphetamine provided comp
	Differentiate the subject in a sentence based on gender	ι endoftext ι -ι M Fundamental FG Fav
	Return a synonym	Word termOn English meanings
	Translate english to spanish	the ththebb volunt
	Return a country's capital city	Ang Suppose AUTHthe beh Assassins
Sentiment	What is the sentiment expressed by the reviewer for the movie?	Pap Azerb Saiyan Forean Talatar Yemeni IndBloomberg receiveda
	How does the author of the news headline feel?	Fur resultolandgroundur augmented=

Summary Thus Far

- In-context learning:
 - Pre-trained LMs imitated examples provided in their context. (why??)
- It turns out there is a **huge variance** in performance depending on the encoding.
 - The choice of demonstrations, their order, wording, etc.
 - You can treat them as hyper-parameters
 - You should **not** choose these encodings based on the test data.
- Generally, you want to an encoding that **makes your task similar to language modeling** — closer to what is observed during pretraining.